

Robotics for Healthcare

Upper Limb Rehab Robotics

上肢康复机器人

Yu Haoyong

Department of Biomedical Engineering, NUS

bieyhy@nus.edu.sg

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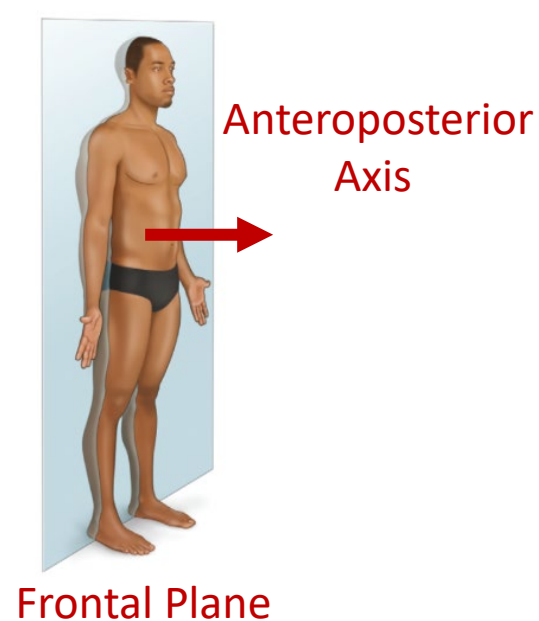
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上肢运动的概念 — 人体解剖三平面

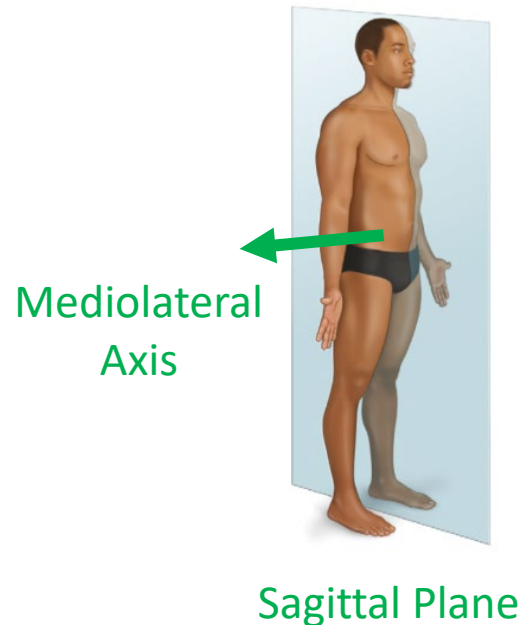
Concepts of upper limb movements

● Human anatomy

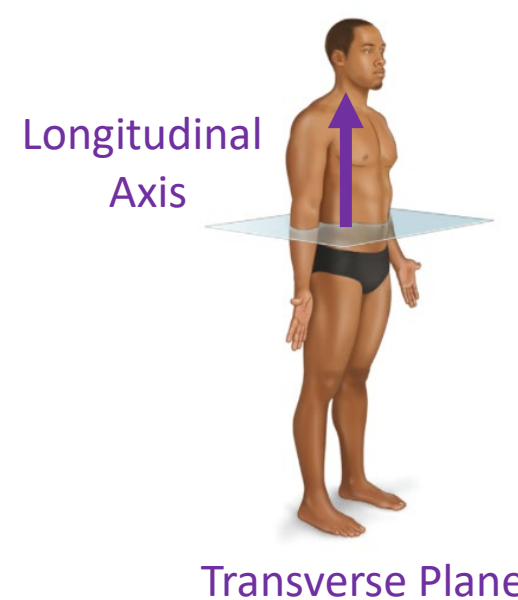
和下肢课件一样，先建立解剖学坐标系。上肢动作大多跨三个平面，不像走路主要在矢状面。



(冠状面/额状面)：矢状轴方向，左右运动



(矢状面)：内外侧轴方向，前后运动



(水平面)：纵轴方向，旋转运动

肩胛骨运动

Concepts of upper limb movements

● Scapula motions

- **Elevation & Depression**

肩胛骨上提/下降（耸肩/沉肩）

- **Retraction & Protraction**

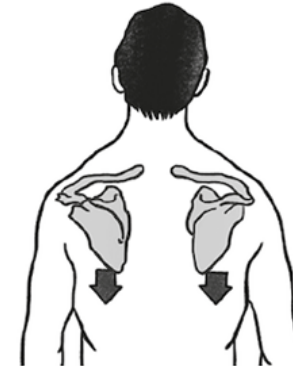
肩胛骨后缩/前伸（夹背/圆肩）

- **Upward rotation & Downward rotation**

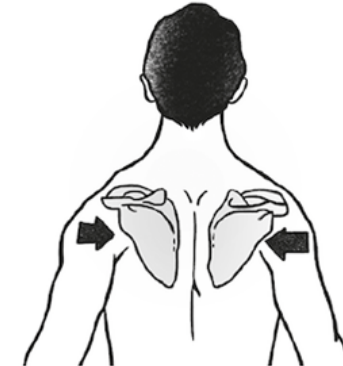
肩胛骨上旋/下旋



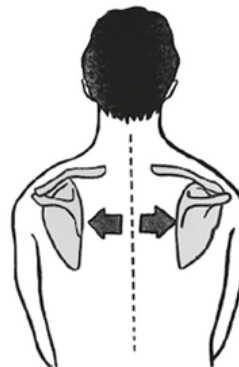
Elevation



Depression



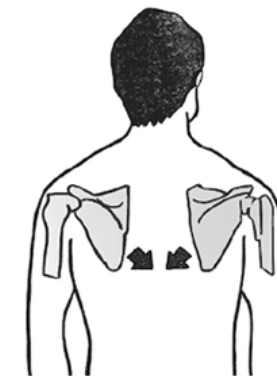
Retraction



Protraction



Upward rotation



Downward rotation

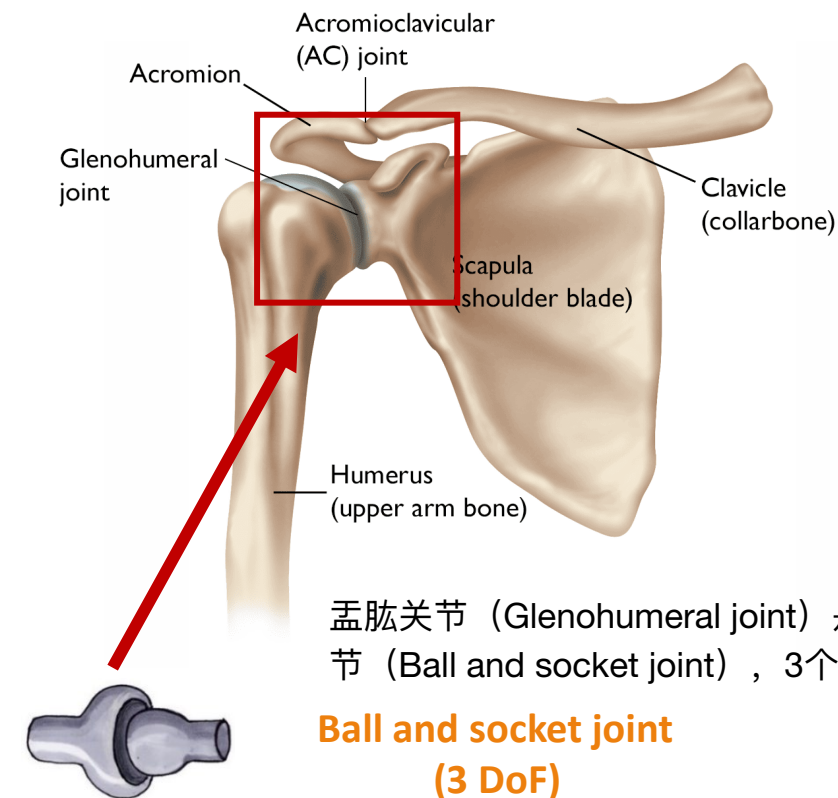
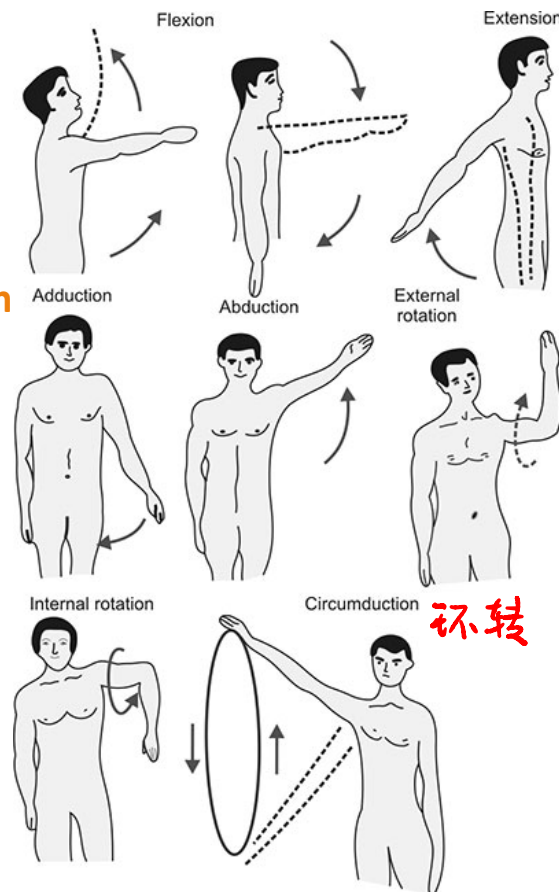
肩胛骨是上肢运动的"底座"。很多外骨骼忽略肩胛骨运动，导致肩关节错位（shoulder misalignment），这是上肢外骨骼设计的一大难点。

肩关节运动

Concepts of upper limb movements

● Shoulder motions

- **Flexion & Extension** 前屈/后伸
- **Abduction & Adduction** 外展/内收
- **Internal rotation & External rotation** 内旋/外旋

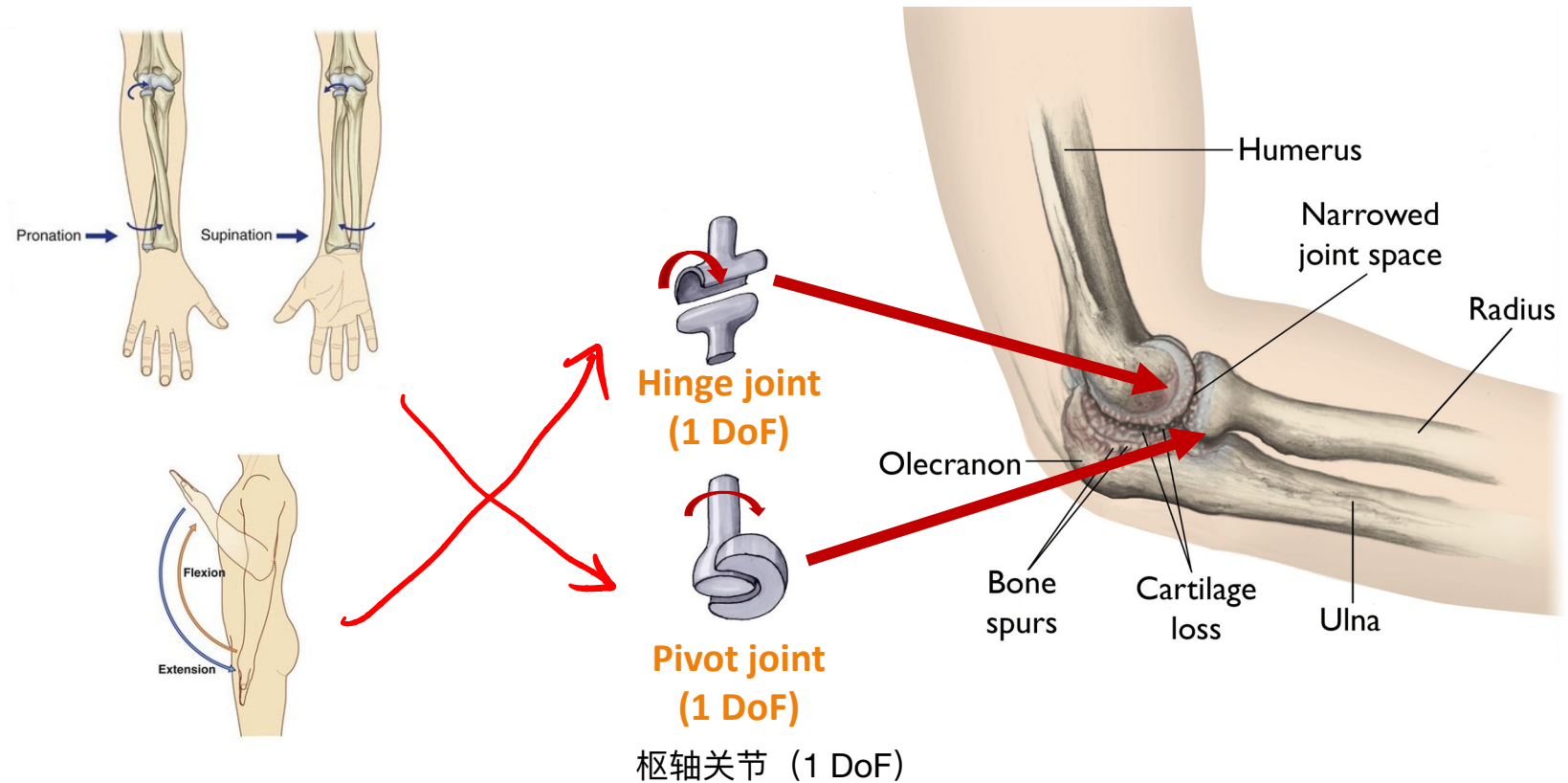


肩关节是全身最灵活的关节，3个DoF。但也是最容易脱位的关节——外骨骼如果强行驱动盂肱关节而不配合肩胛骨运动，会造成肩关节半脱位（subluxation）。

肘关节运动 Concepts of upper limb movements

● Elbow motions

- **Pronation & Supination**
前臂旋前/旋后（手心朝上/朝下）
- **Flexion & Extension**
肘屈曲/伸展——铰链关节



肘关节实际是两个独立关节的组合——
肱尺关节管屈伸，上尺桡关节管旋转。
所以肘部一共2个DoF。

腕关节

Concepts of upper limb movements

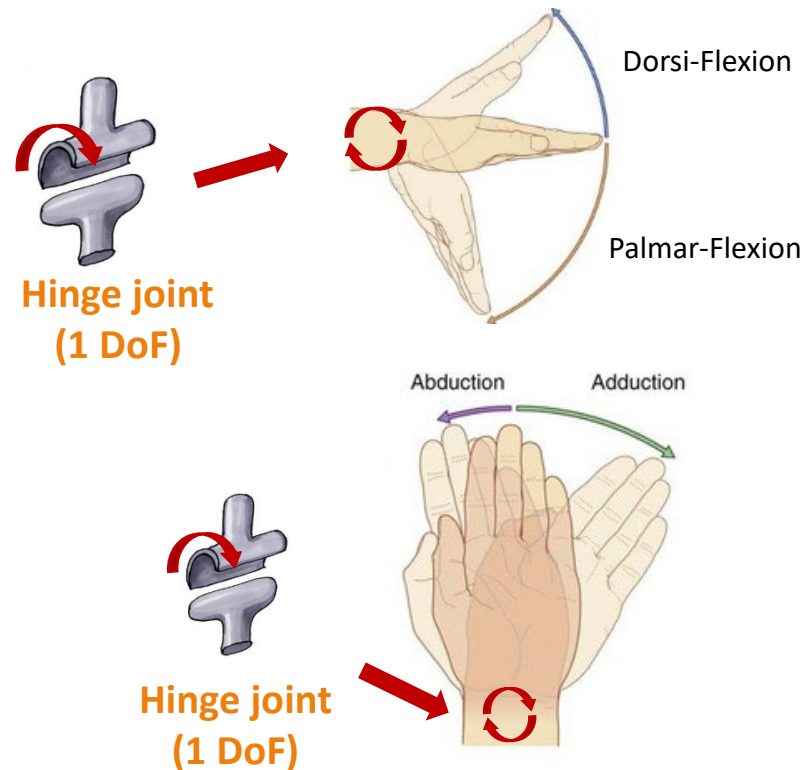
● Wrist motions

- **Dorsi-Flexion & Palmar-Flexion**

腕背屈/掌屈（勾手腕/压手腕）——铰链关节1 DoF

- **Abduction & Adduction**

腕桡偏/尺偏（向拇指侧/小指侧弯）——另一个铰链关节1 DoF



腕关节2个DoF，都是铰链型，结构相对简单，所以RiceWrist这类早期专门做腕关节康复的机器人很多。

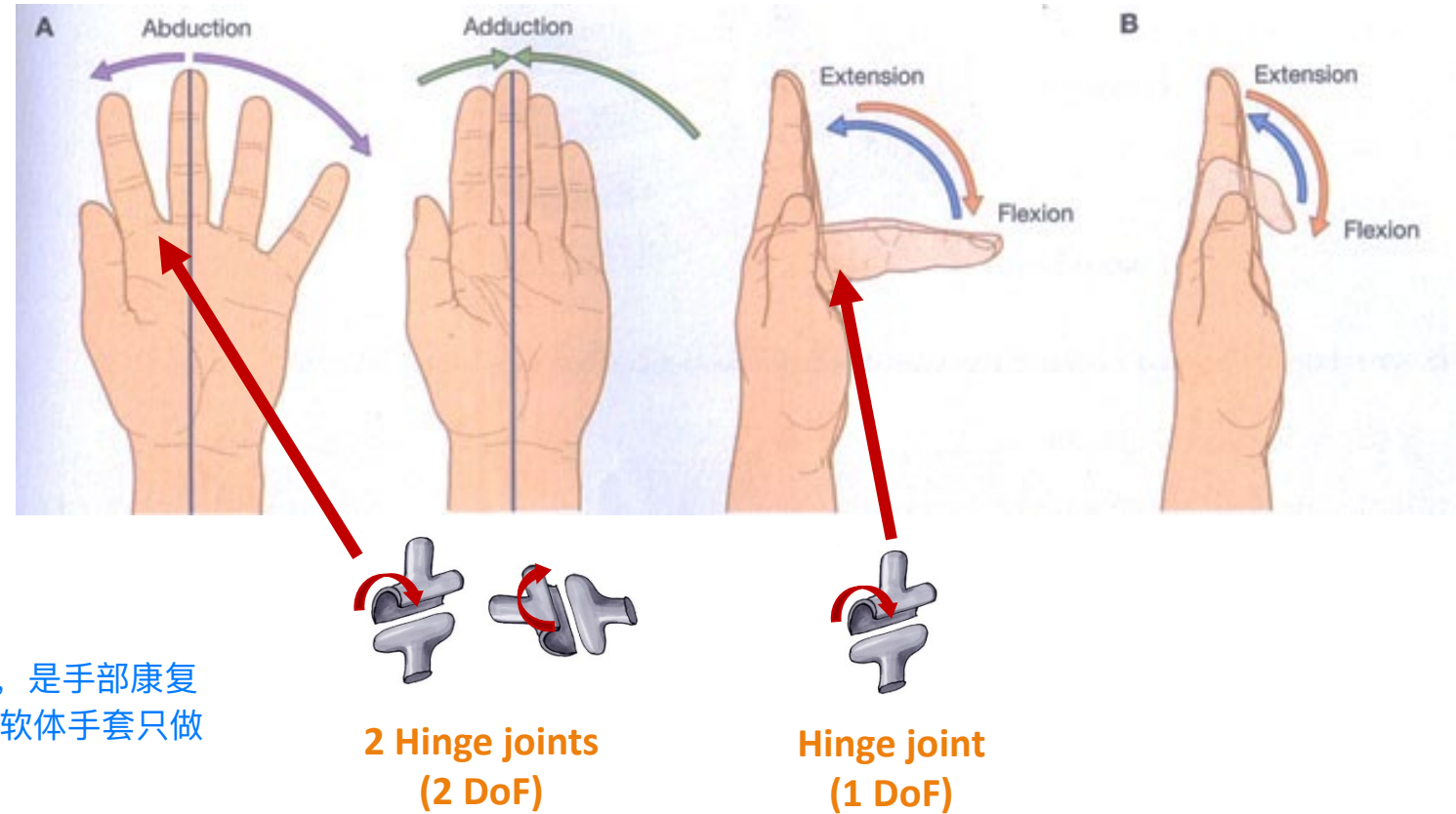
手指运动

Concepts of upper limb movements

● Finger motions

- **Flexion & Extension**
手指屈伸
- **Abduction & Adduction**
手指外展/内收 (张开/并拢)

MCP关节 (2 DoF, 铰链+侧摆) + PIP/DIP关节 (各1 DoF铰链)。



手指自由度极多 (单手约20+ DoF)，是手部康复机器人最难做的部分——所以大多用软体手套只做屈伸，不管侧摆。

中风后上肢功能的故障 Upper-limb Impairments after Stroke

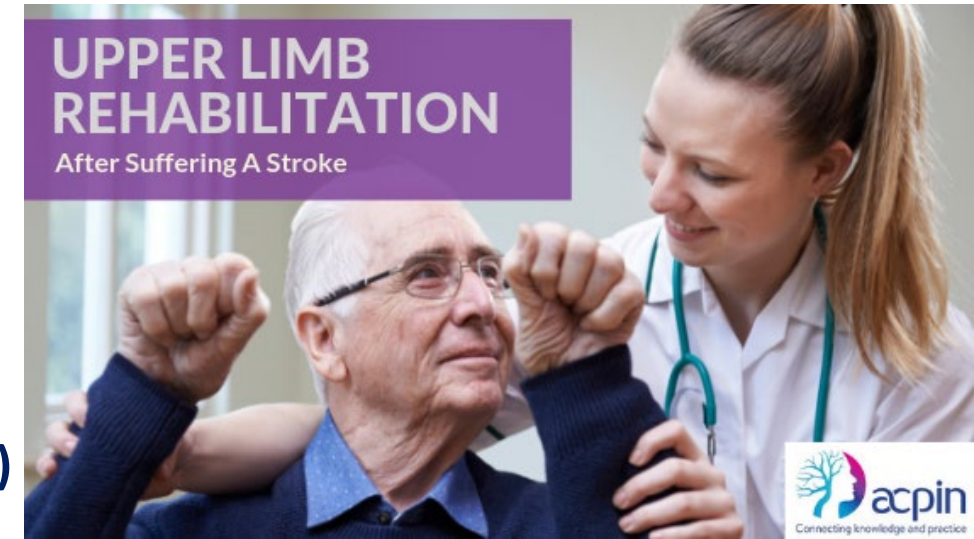


- Stroke patients lose motor functions due to brain damage and absence of connection between the brain and body
 - Patients suffer various physical sequelae
- 核心：中风患者因脑损伤、脑-体连接中断而丧失运动功能，遗留各种躯体后遗症。

上肢康复目标

Goal of upper limb rehab

- **Facilitates neuroplasticity to reconnect neural pathways and synapses & improve functional outcomes**
 - 促进神经可塑性，重建神经通路突触，改善功能结局
- **Reduce spasticity**
 - 降低痉挛
- **Inhibit atypical muscle synergies**
 - 抑制异常肌肉协同
- **Prevent muscle contractures**
 - 预防肌肉挛缩
- **Avoid learned nonuse (e.g., Use healthy limb to do everything)**
 - 避免习得性废用（即"凡事都用好手"）
- **Relearn interlimb coordination**
 - 重新学习双侧肢体协调



和中风课件CIMT那一节呼应——中风患者会自发只用健手，时间长了患手真的"被大脑放弃"。

实现恢复阶段 (Brunnstrom 模型)

Approach of upper limb rehab

● Brunnstrom Approach

Stage 1: Flaccid paralysis (软瘫) → 做被动关节活动度练习

Stage 2: Spasticity appears (痉挛开始出现)

Stage 3: Spasticity increased (痉挛加重)

Stage 4: Spasticity decreased (痉挛开始下降)

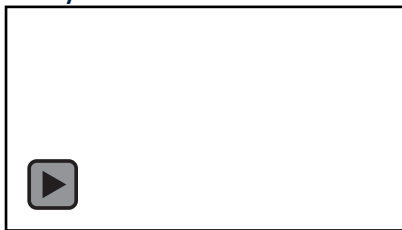
Stage 5: Complex movement (复杂运动出现)

Stage 6: Spasticity disappears (痉挛消失)

Stage 7: Normal (正常)



Stage 1:
Flaccid
paralysis



(Helen Hayes Hospital, 2018)

Stage 2:
Spasticity
(Appear)

Stage 3:
Spasticity
(Increased)

Stage 4:
Spasticity
(Decreased)

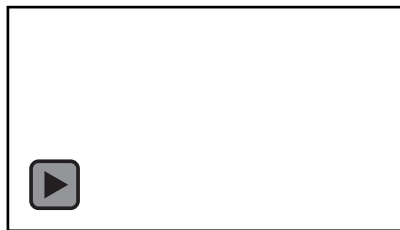
Stage 5:
Complex
Movement

Stage 6:
Spasticity
(Disappear)

Stage 7:
Normal

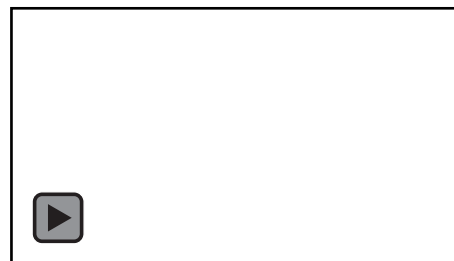


Active Range of Motion Exercise



(Speech Therapy Practice, 2018)

4-6阶段做主动抗阻训练+ADL训练。
1) Active Range of Motion Exercise with resistance
2) Activities of daily living (ADL) Training



(Helen Hayes Hospital, 2016)

Passive Range of
Motion Exercise

Brunnstrom是中风—运动恢复的经典分期模型——先软瘫、再痉挛、痉挛先升后降、最后恢复正常。康复机器人要根据患者处于哪个阶段选不同训练模式。

ADL训练 (Activities of Daily Living) Approach of upper limb rehab

● ADL Training



• Basic ADLs: 进食、洗澡、穿衣、转移、如厕、行走移动



工具性日常生活：做饭、打扫、服药、洗衣、购物、理财、打电话、乘车

Task-specific



Unilateral



Bilateral: Symmetric



Bilateral: Asymmetric

任务难度递进：单侧 → 双侧对称 → 双侧不对称，从简单到难。

Simple to difficult tasks

ADL是康复的终极目标——不是让患者空挥手臂，而是能自己吃饭穿衣出门。机器人做的游戏训练必须最终迁移到真实ADL。

Reference from New Life Styles, <https://www.newlifestyles.com/blog/adls-iadls-what-are-they-how-do-you-measure-them-why-might-storytelling-help-those-who-are-adl-deficient>

Adapted from Yuvakrishna Developmental Hub, <http://www.yuvakrishna.in/activities-of-daily-life-training.php>

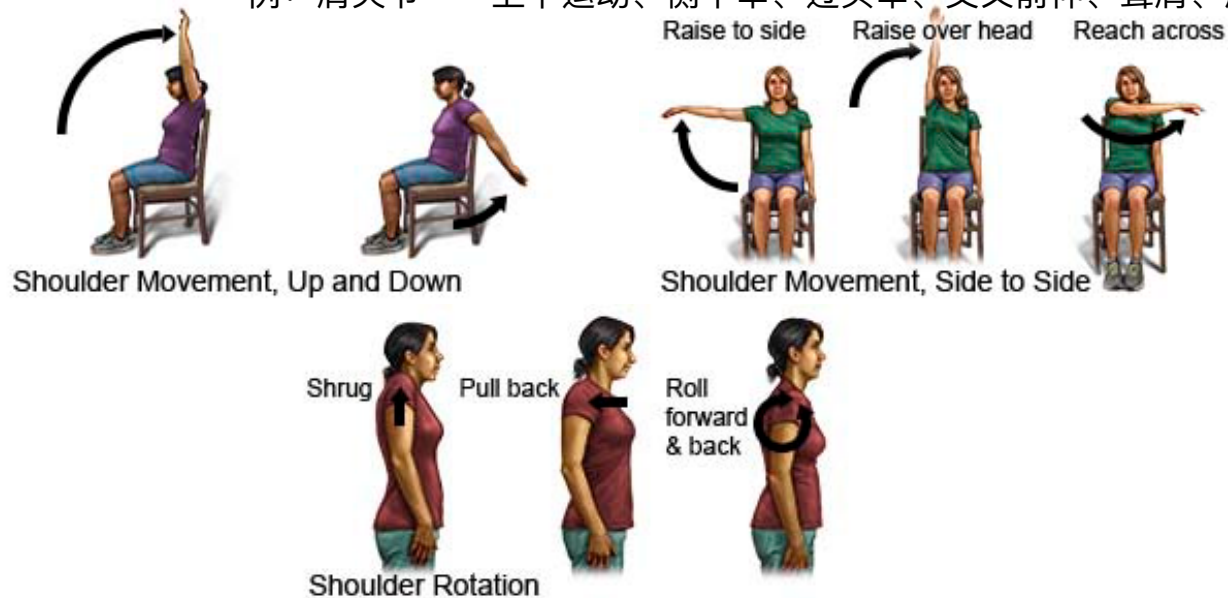
Adapted from eugeria, <https://info.eugeria.ca/en/what-are-iadls-and-how-are-they-different-from-adls/>

Adapted from CAHAI, <http://www.cahai.ca/layout/content/CAHAI-Manual-Singapore.pdf>

上肢康复训练 —— 主动/被动关节活动度练习

Upper Limb Rehab Training Schemes

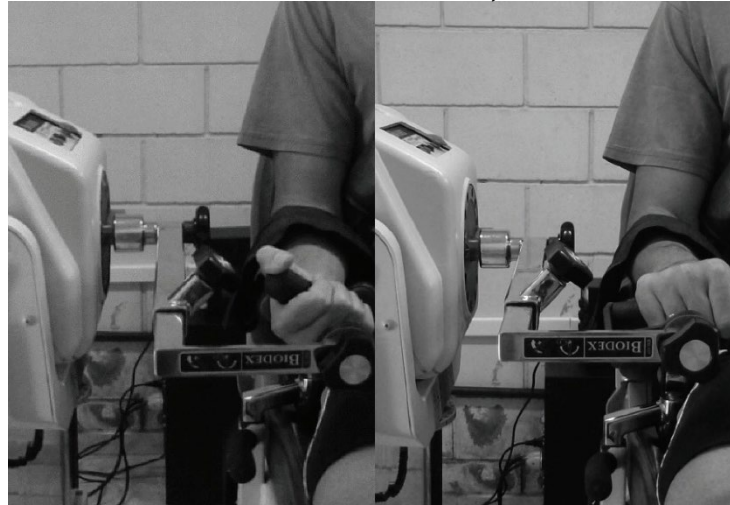
- Active/Passive **Range of Motion (ROM)** Exercise
 - Reduce pain, increase joint flexibility, recover motor synergies
 - Actively move the specific joint in various direction
 - E.g. Shoulder
 - 目的：减痛、增加关节柔韧度、恢复运动协同
 - 主动运动特定关节各个方向
 - 例：肩关节——上下运动、侧平举、过头举、交叉前伸、耸肩、后拉、前后绕环



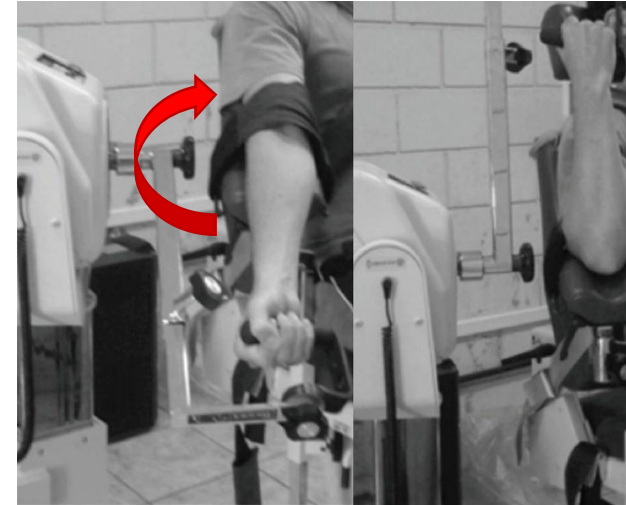
Adapted from Drugs.com, <https://www.drugs.com/cg/active-range-of-motion-exercises.html>

力量训练 (阻抗主动关节活动) Upper Limb Rehab Training Schemes

- Strengthening Training (AROM with resistance)
 - Improve muscle weakness(Hemiparesis) • 改善肌无力 (偏瘫)
 - Isometric: Maintain the muscle strength without movement
 - Isokinetic: Maximize the muscle strength with constant-speed movement
 - 等长训练 (Isometric) : 关节不动, 肌肉静态收缩维持力量
 - 等速训练 (Isokinetic) : 恒定速度运动, 最大化肌肉力量



Isometric training for elbow flexor(L)
and extensor(R) muscle



Isokinetic training for elbow flexor muscle:
Starting position(L) and Ending position(R).

等速肌力测试仪
(Biodex类设备) 测
肘屈肌/伸肌。

Adapted from N. M. Bassan *et al.*, "Reliability of isometric and isokinetic peak torque of elbow flexors and elbow extensors muscles in trained swimmers," *Brazilian Journal of Kinanthropometry and Human Performance*; Vol 17, No 5 (2015), 2015.

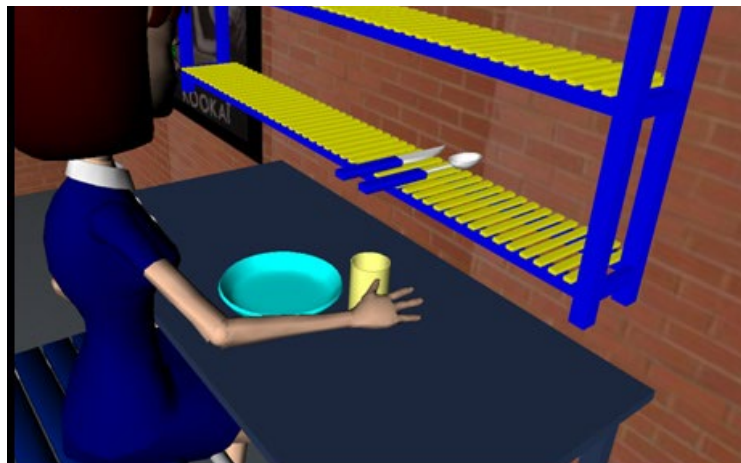
等长=肌肉收缩但长度不变 (推墙) ; 等速=设备强制你以恒定角速度运动, 无论你多大力。这是康复评估和肌力训练的金标准设备。

Upper Limb Rehab Training Schemes

- Activities of daily living (ADL) Training
 - Task-Specific (Important and meaningful task)
 - Recover motor synergies and fine motor skill
 - More challenging to achieve with a robotic device
 - Require large range of motions (RoM), many degrees of freedom (DoF) and sophisticated control algorithms
- 任务特异性（重要且有意义的任务）
 - 恢复运动协同和精细运动技能
 - 用机器人做ADL挑战更大：需要大活动范围、多自由度、复杂控制算法



ARMin外骨骼 ARMin



Graphical display for the ARMin's ADL training
ARMin的虚拟ADL训练场景（虚拟厨房拿杯子）

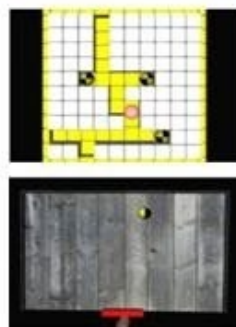
Adapted from T. Nef, V. Klamroth-Marganska, and R. Riener, "ARMin - Exoskeleton Robot for Stroke Rehabilitation," vol. 25, 2010, pp. 127-130.

上肢康复机器人的作用，案例研究 I — ARMin II

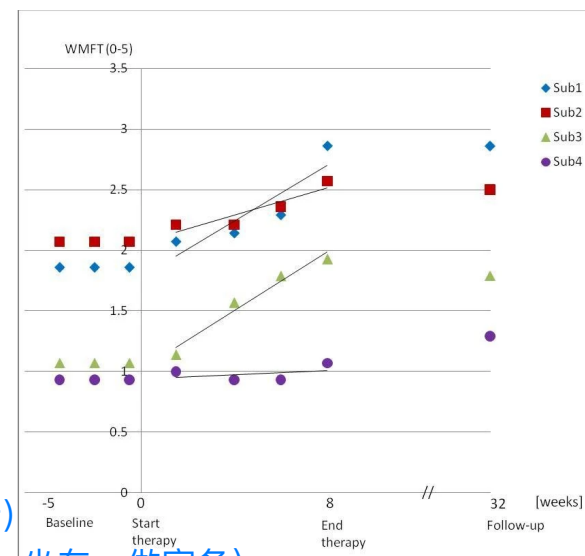
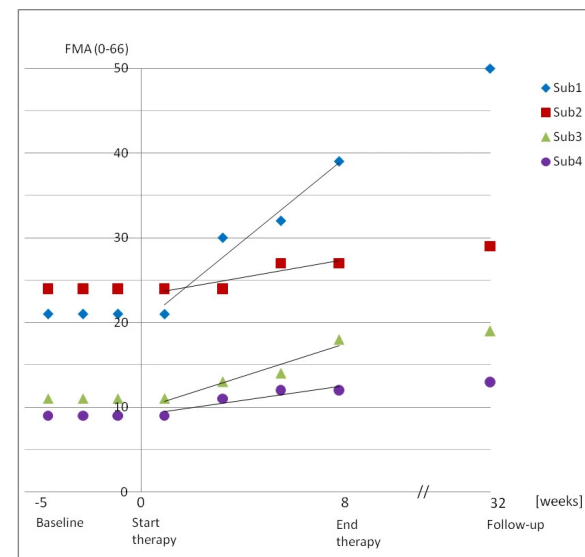
Effects of upper rehab robotics

● Case study 1: ARMin II rehab robotics

ADL(Activities of Daily Living)即日常生活活动，分为基础性日常生活活动BADL与工具性日常生活活动IADL。BADL指进食、穿衣、如厕等自我照料基础活动；IADL指做饭、购物、理财等社会独立生活所需要的高阶任务。中风康复提倡任务特异性ADL训练，通过完成真实生活任务促进神经可塑性，提升患者独立生活能力。



- Fugl-Meyer Assessment (FMA) assesses the sensorimotor impairment in post-stroke patient
- Total score, maximum = 66
Score for shoulder/elbow, max. = 36
Score for wrist/hand, max. = 30
评估量表：
 - FMA: 评估中风患者感觉运动缺损，总分66（肩/肘36分，腕/手30分）
 - WMFT: 通过计时功能性任务评估上肢运动能力，正常=5分
- Wolf Motor Function Test (WMFT) assesses upper extremity motor ability through timed and functional tasks
- Normal motor functions = 5
- BADL: 管好自己身体（吃、穿、洗澡、上厕所）
- IADL: 管好自己的生活环境（买菜做饭、花钱、坐车、做家务）



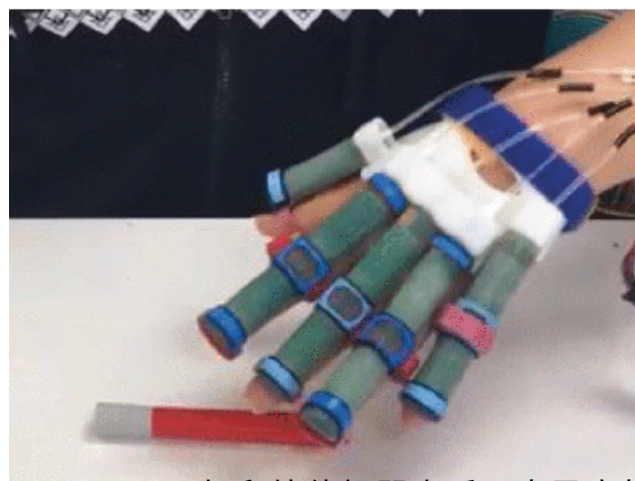
案例12 - 软件机器人手

Effects of upper rehab robotics

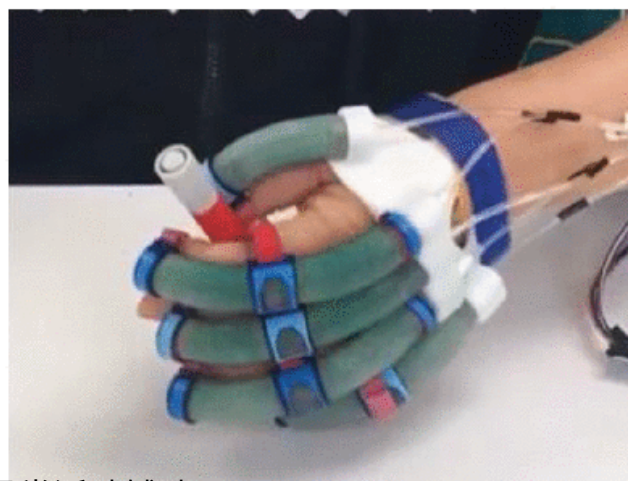
● Case study 2: Soft robotic hand (Heung et al., 2019)

- 对比训练前后：a) 桌上抓物
b) 手指独立屈伸，均改善

[Opening]



[Grasping]



- 3D打印软体机器人手，中风康复+日常活动辅助
- 患者戴上后可完成：开手、抓笔、做"胜利"手势
- Observed on performing daily tasks (ADL), e.g. grasping a pen, and performing a win gesture

a) [Before]



[After]



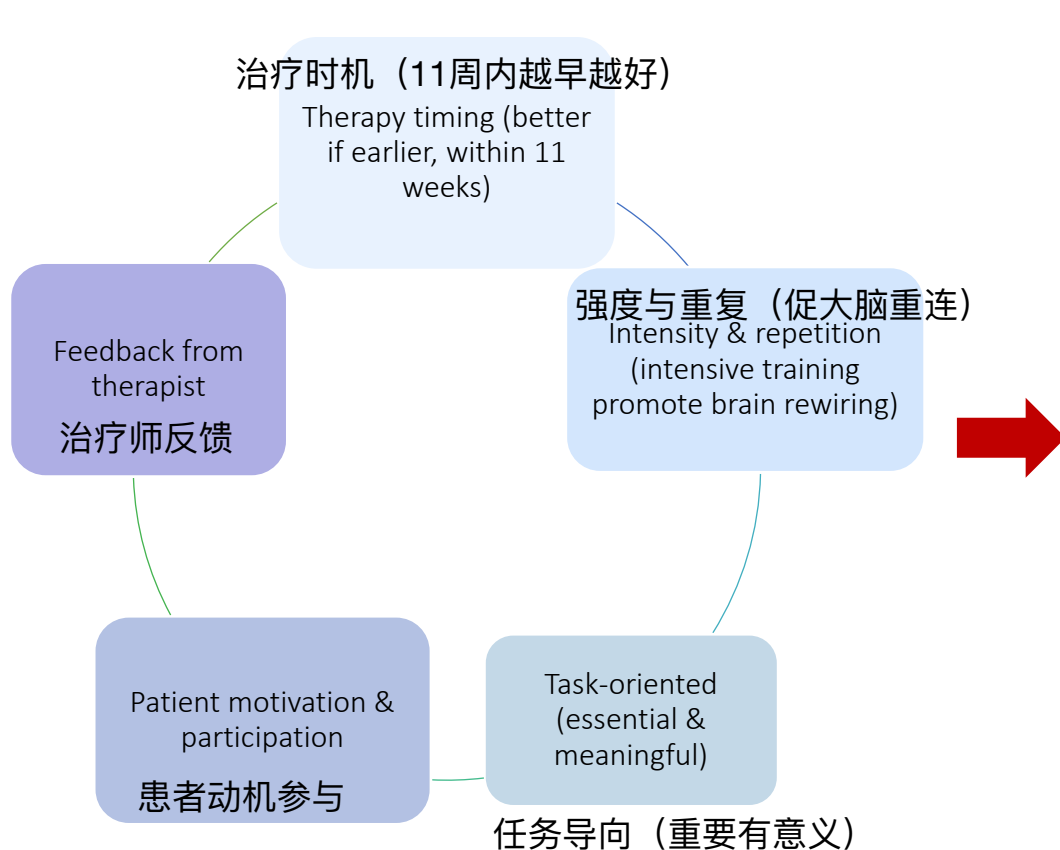
b)



康复机器人优势

Advantages of Rehab Robotics

神经康复要素闭环



机器人辅助治疗 (RT) 四大优势: Robot-assisted Therapy (RT)

High-dosage, high-intensity training
1. 高剂量、高强度训练

Highly consistent training
2. 高度一致的训练

Reduce burden on therapists
3. 减轻治疗师负担

Quantitative Evaluation on patients' recovery progress
4. 对患者康复进展的定量评估

上肢康复机器人分类

Categories of upper limb rehab robotics

上肢康复机器人发展史

History of upper limb rehab robotics

● MIT-MANUS (1990s)

MIT-MANUS (1990s)

世界第一台上肢康复机器人。MIT-Manus设计用来帮助中风幸存者恢复手臂运动，外骨骼支撑患者手臂完成一系列程序化练习。



(MIT-Manus Robot)

MIT-Manus就是现在InMotion Arm Robot的前身，由MIT的Hogan和Krebs教授团队开发，开创了整个康复机器人领域。

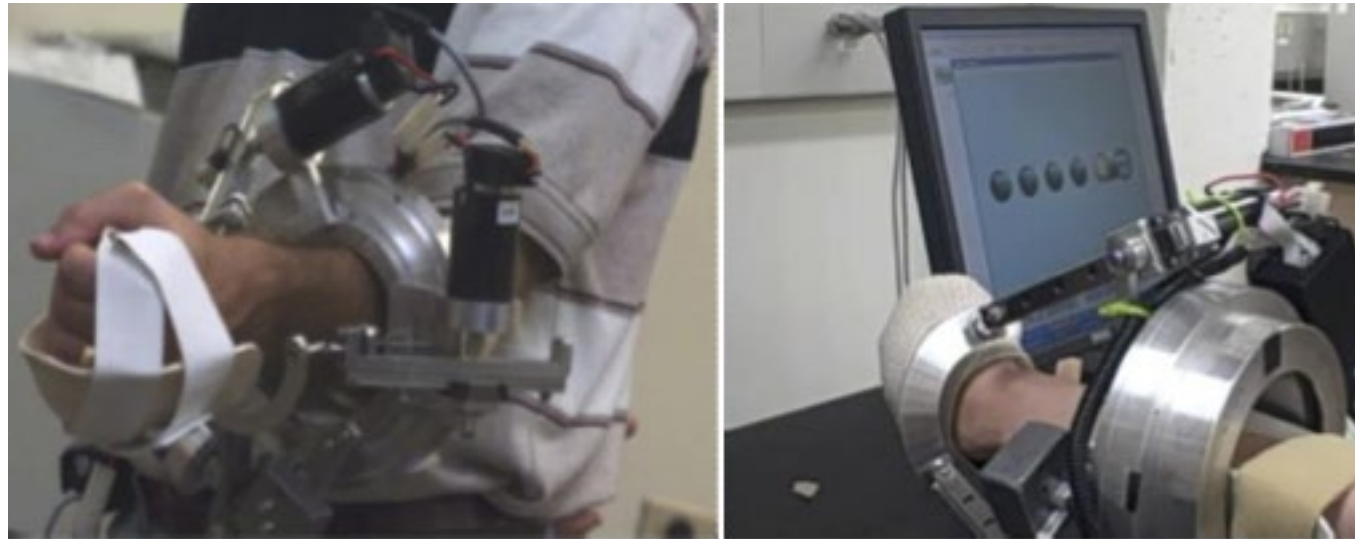
- The first upper limb rehabilitation robot. MIT-MANUS was designed to aid stroke survivors in regaining arm movement and featured an exoskeleton that supported the patient's arm through a series of programmed exercises.

History of upper limb rehab robotics

● RiceWrist (1997)

RiceWrist (1997)

另一台早期上肢康复机器人，聚焦腕关节康复，提供受控运动，带力反馈。



(RiceWrist,1997)

- The RiceWrist was another early upper limb rehabilitation robot developed in the late 1990s. It focused on wrist rehabilitation and provided controlled movement for patients.

History of upper limb rehab robotics

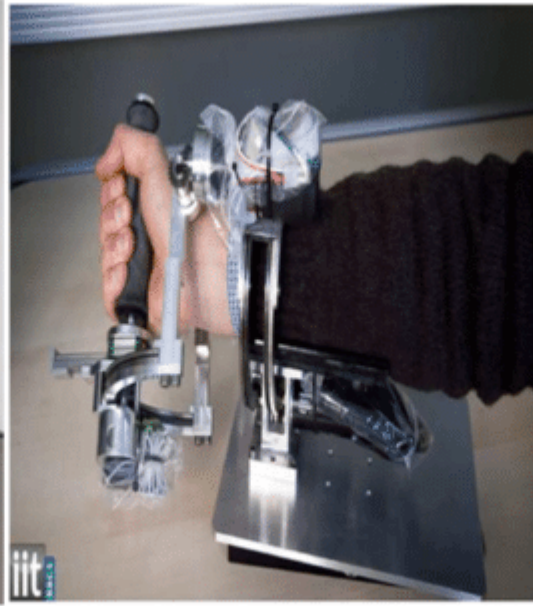
外骨骼与末端执行器并行 (2000s)

● Exoskeletons and End-Effector Devices (2000s)

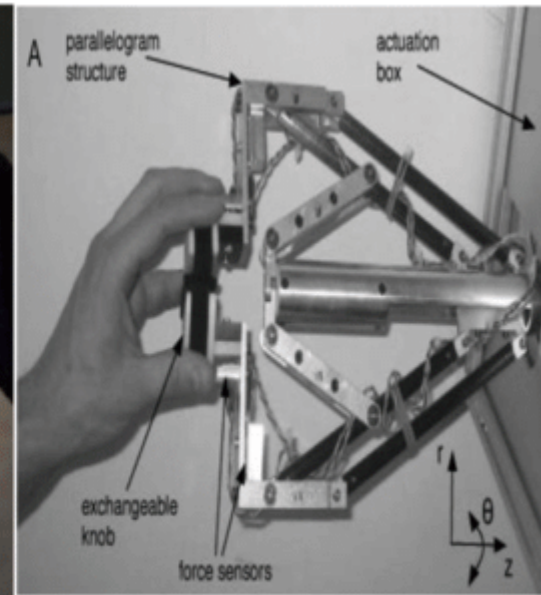
2000年代开始转向外骨骼——包裹肢体提供支撑和运动辅助。研究者同时探索两条路线：



(a) 外骨骼：包裹整个手臂



(b)



(c)

• 末端执行器：只与肢体末端（手）交互

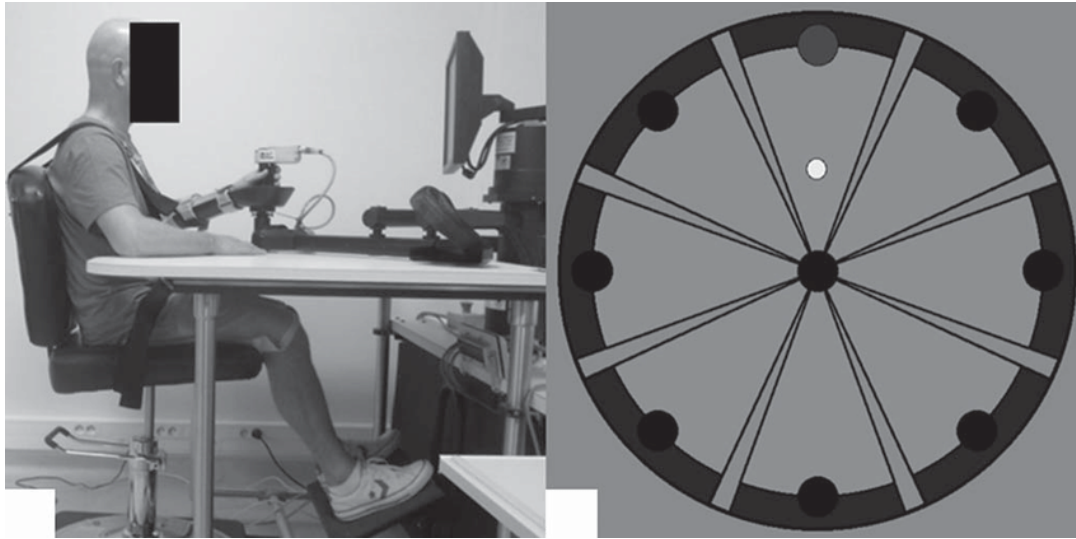
- The 2000s saw a shift towards the development of exoskeletons, wearable robotic devices that encompass the limb to provide support and movement assistance. Researchers began exploring both exoskeletons and end-effector devices, which interact with the end (hand) of the limb rather than the entire limb.

History of upper limb rehab robotics

临床商业化 (2000s至今)

● Rehabilitation Robotics in Clinical Settings (2000s-Present)

- InMotion Arm Robot (末端执行器, 带游戏化环形轨迹任务)。



(InMotion Arm Robot)

- ArmeoSpring (被动重力补偿悬吊臂)



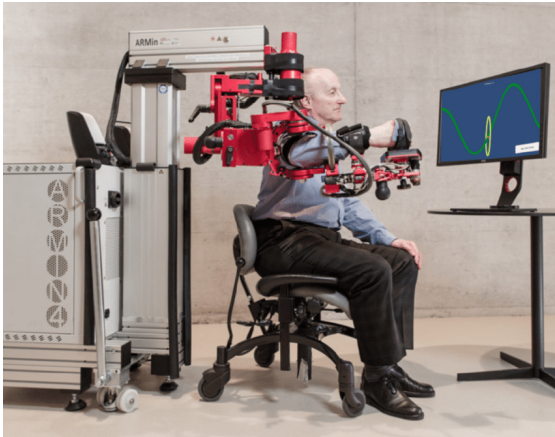
(ArmeoSpring Robot)

- As technology improved, rehabilitation robots started making their way into clinical settings for broader use.
- Robotic devices like the InMotion Arm Robot and the ArmeoSpring became commercially available for rehabilitation purposes.

History of upper limb rehab robotics

传感与控制进步 (2010s至今)

● Advancements in Sensing and Control (2010s-Present)



(Armin IV)



(UCLH stroke rehab robot)

ARmin IV、UCLH中风康复机器人



- In recent years, there has been a focus on incorporating advanced sensing technologies, such as electromyography (EMG) and force sensors, to enhance the control and adaptability of robotic systems.
 - Machine learning and artificial intelligence have also been integrated to personalize rehabilitation programs based on individual patient needs and progress.
- 近年，主要聚焦于融入EMG肌电、力传感器等高级传感技术。 • 机器学习和AI用于个性化康复方案

Armin IV: <https://www.armin-robot.com/en/homePage>

UCLH <https://www.uclhroboticsfellowship.com/>

History of upper limb rehab robotics

● Widespread Adoption and Diversity of Devices (Present) 设备多样化与家庭化 (现在)



(ETH ANYexo II)



(Yeecon Rehab Robot)



(Armeo®Power - Hocoma)

- Today, a variety of upper limb rehabilitation robotic devices are available, ranging from exoskeletons to end-effector devices.
- These devices are used not only in clinical settings but also in home-based rehabilitation programs, providing patients with increased access to ongoing therapy.

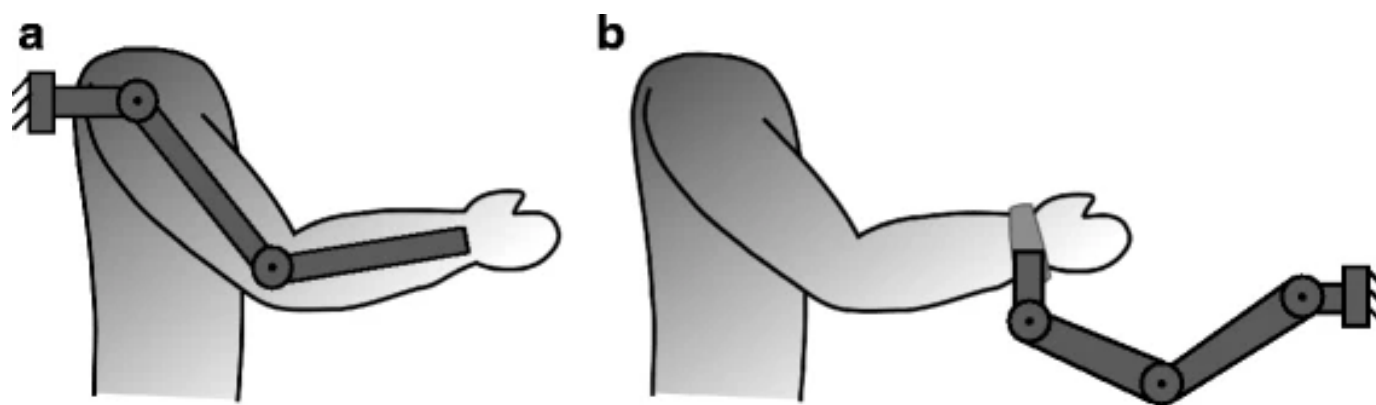
- 现在从外骨骼到末端执行器各种设备都有
- 不仅用于临床，还用于居家康复，让患者更容易持续接受治疗

Categories of upper limb rehab robotics

末端执行器 vs 外骨骼

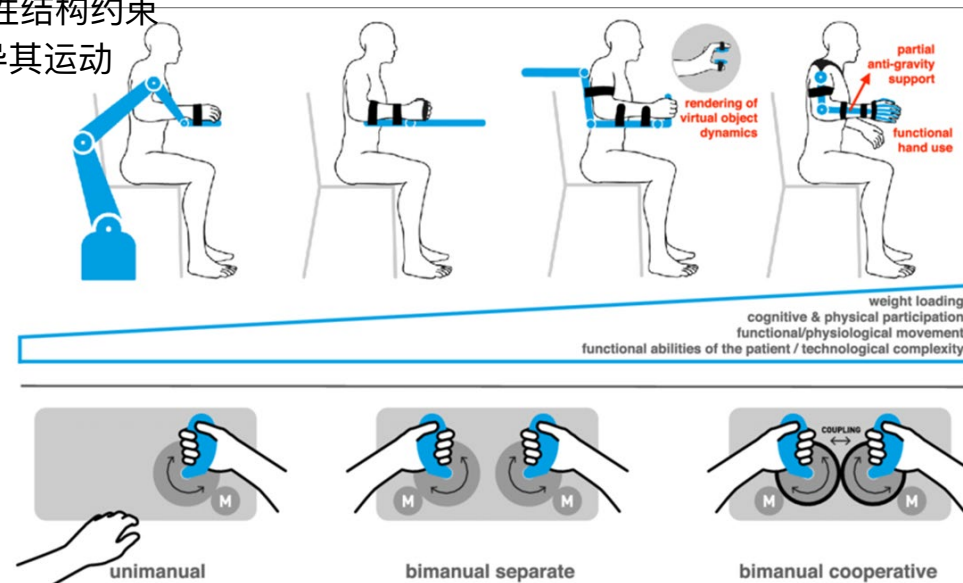
● End-effector-based vs Exoskeleton-based

- 末端执行器：与肢体末端（手/腕）交互，允许末端在三维空间自由运动，患者手臂通常不被刚性结构约束
- 外骨骼：可穿戴机器人结构，包裹并支撑肢体，模仿人体骨骼和关节结构，直接支撑肢体并引导其运动



(Exoskeleton-based)

(End-effector)



- End-effector devices **interact with the end** (typically the hand or wrist) of the limb. They allow movement of the end-effector in three-dimensional space, and the patient's arm is usually not constrained by a rigid structure.
- Exoskeletons are wearable robotic structures that **encompass and support the limb**. They mimic the skeletal structure and joints of the human body, providing direct support to the limb and guiding its movement.

Reference from P. Maciejasz, J. Eschweiler, K. Gerlach-Hahn, A. Jansen-Troy, and S. Leonhardt, "A survey on robotic devices for upper limb rehabilitation," *Journal of NeuroEngineering and Rehabilitation*, vol. 11, no. 1, p. 3, 2014.

Adapted from S. Micera *et al.*, "A Simple Robotic System for Neurorehabilitation," *Autonomous Robots*, vol. 19, no. 3, p. 271, 2005.

Crocher V, Fong J, Bosch T J, et al. Upper limb dewatering using underactuated end-effector-based backdrivable manipulanda[J]. IEEE Robotics and Automation Letters, 2018, 3(3): 2116-2122.

末端执行器型 — 实例

Categories of upper limb rehab robotics

● End-effector based

- 优点：更适应不同臂长臂形；对患者不笨重；比外骨骼成本低
- 缺点：多自由度情况下，肢体位置控制更复杂（只抓手，肘肩角度不确定）



(InMotion Arm Robot)



(MEMOS)



[RAL, 2018]

- Advantages: more adaptable to various arm sizes and shapes; less cumbersome for patients; more cost-effective compared to exoskeletons.
- Disadvantages: the control of the limb position may be more complicated, in the case with multiple degrees of freedom

Categories of upper limb rehab robotics

- End-effector based



(InMotion Arm Robot)

外骨骼型 - 实例

Categories of upper limb rehab robotics

● Exoskeleton-based

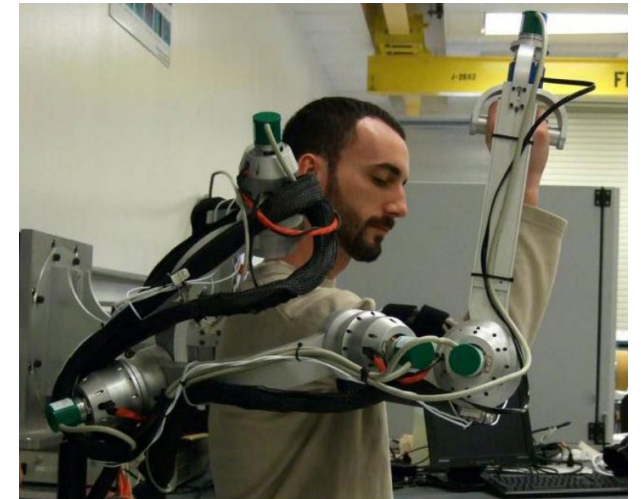
- 优点：对肢体运动控制更直接；能对特定关节提供精准靶向支撑；适合重度损伤患者
- 缺点：结构比末端执行器复杂



(Yeecon Rehab Robot)



(ETH ANYexo II)



(MGA Exoskeleton)

- Advantages: provides more direct control over limb movement; offers precise and targeted support to specific joints; can be suitable for patients with severe impairments.
- Disadvantages: more complicated device structure than an end-effector based device

Categories of upper limb rehab robotics

- Examples of Exoskeleton-based



(InMotion Arm Robot)

上肢康复机器人的设计 — 设计原则

Design of upper limb rehab robotics

● Design principles

上肢各关节活动范围 (RoM)

The values of RoM (range of motion) of upper limbs

Articulation	Movement	Range of motion [°]
Shoulder	Extension - Flexion	50–180
	Abduction- Adduction	180–30
	Internal – external rotation	100–80
Elbow	Extension-Flexion	0–145
	Pronation- Supination	85–90
Wrist	Flexion -Extension	85–85
	Adduction- Abduction	45–15

四大设计原则：

1. 以患者为中心：理解目标患者群需求和能力。

2. 生物力学适配：提供足够RoM和DoF满足特定康复训练

3. 轻量化人体工学：成本可控，可在康复中心广泛采用。

4. 可调节可定制：适配不同体型、活动范围、肌力水平

• **Patient-Centered Design:** Understand the needs and capabilities of the target patient population

• **Biomechanical Compatibility:** provide a sufficient range of motion and degrees of freedom to address specific rehabilitation exercises.

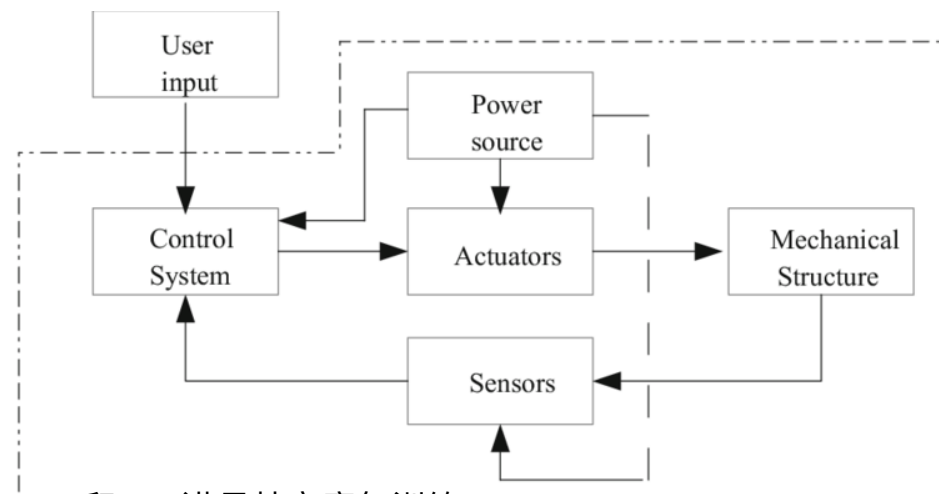
• **Lightweight and ergonomic design:** strive to create a cost-effective solution that can be widely adopted in rehabilitation centers and clinics.

• **Adjustability and Customization:** provide adjustable settings to accommodate various patient sizes, range of motion, and strength levels..

- 肩：屈伸50—180°，外展内收180—30°，内外旋100—80°
- 肘：屈伸0—145°，旋前旋后85—90°
- 腕：屈伸85—85°，尺偏桡偏45—15°

康复机器人基本模块——用户输入→控制系统→驱动器→机械结构→传感器→反馈回路，电源供电。

Basic Modules of a rehab robot

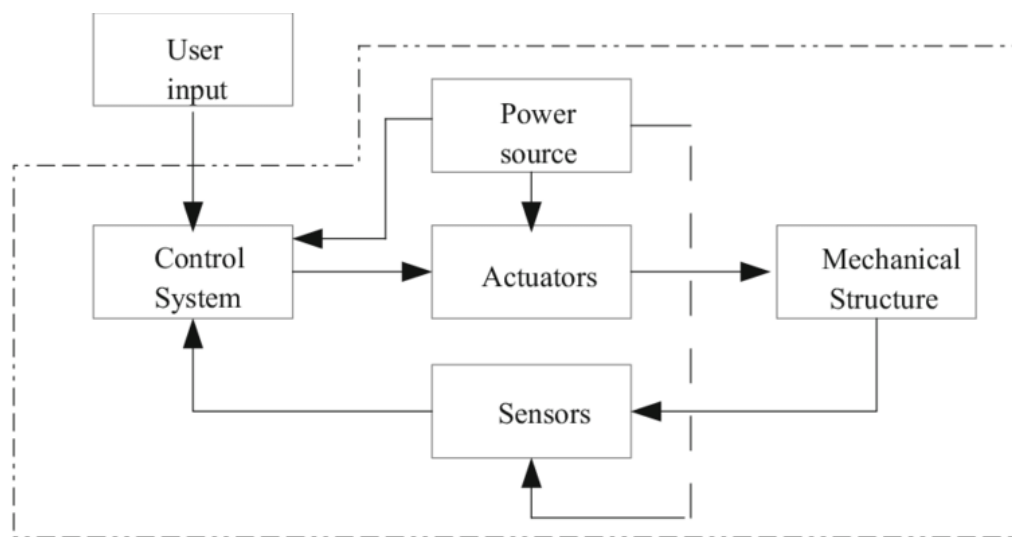


基本组成部件

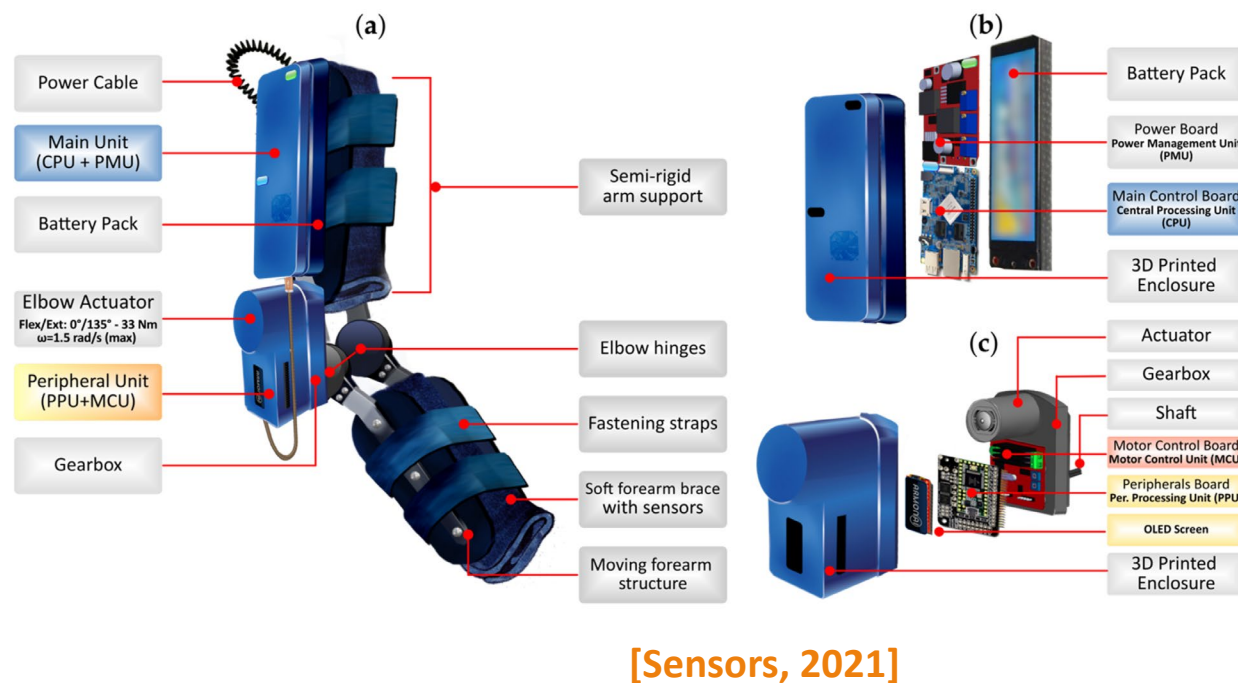
Design of upper limb rehab robotics

● Basic components

右图：一款智能可穿戴上肢外骨骼实物拆解——主单元CPU+PMU、电池包、肘驱动器、外围单元PPU+MCU、齿轮箱、3D打印外壳、电机、力传感器等。



[Applied Sciences, 2020]



[Sensors, 2021]

- Upper limb rehab robotics consist of mechanical structures, **joint actuators**, **batteries**, and a series of **sensors**. Actuators are the most important part. 上肢康复机器人由机械结构、关节驱动器、电池、一系列传感器组成。驱动器是最重要的部分。

Qassim H M, Wan Hasan W Z. A review on upper limb rehabilitation robots[J]. Applied Sciences, 2020, 10(19): 6976.

Vélez-Guerrero M A, Callejas-Cuervo M, Mazzoleni S. Design, Development, and Testing of an Intelligent Wearable Robotic Exoskeleton Prototype for Upper Limb Rehabilitation[J]. Sensors, 2021, 21(16): 5411.

机械结构

Design of upper limb rehab robotics

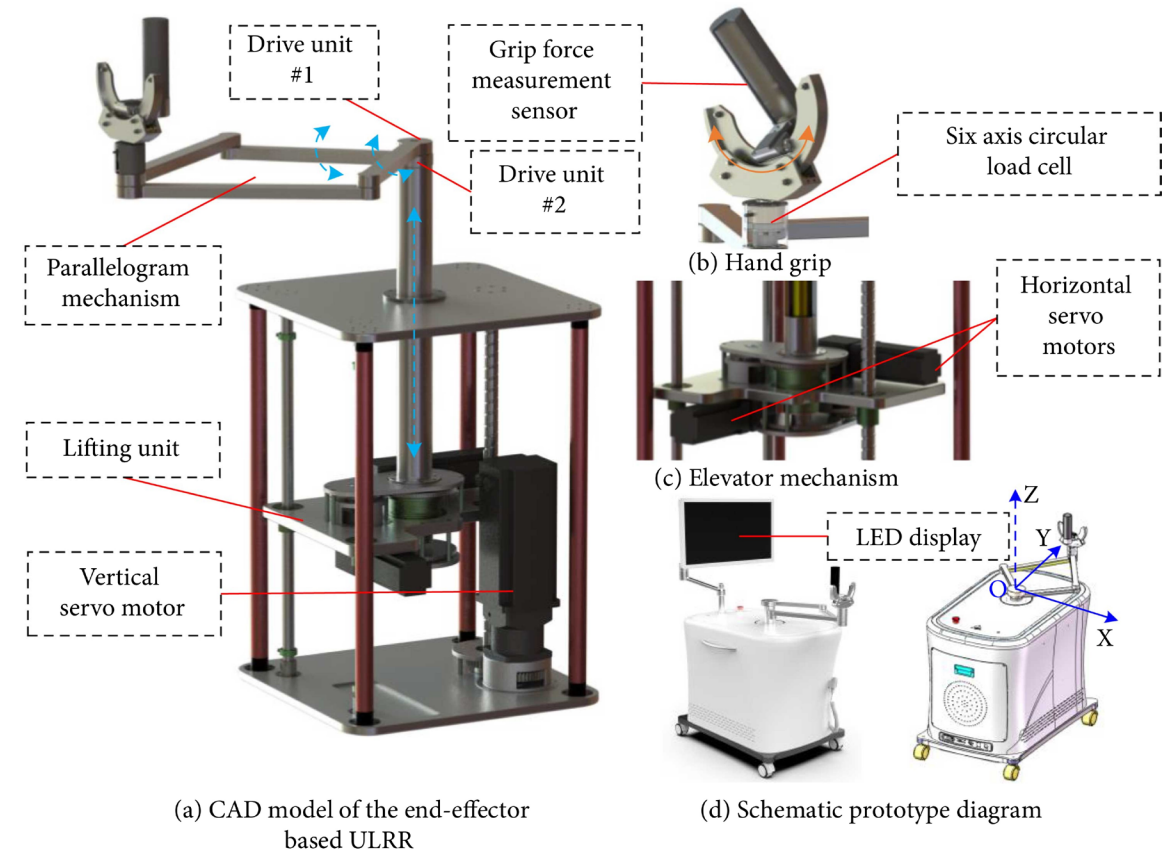
末端执行器CAD——平行四边形机构、驱动单元、握力传感器、六维力传感器、升降机构。

● Mechanical structures

手attachment——可交换手柄、末端执行器、绑带、快换手指固定。



仿人体骨架、关节结构、柔性attachment。外骨骼组件包括支撑部分、缓冲垫、绑带、伸缩杆、承重杆。



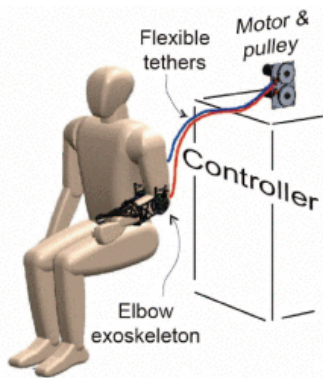
- **Mechanical components:** human body skeleton structures, joint structures, and a flexible attachment. The different components of the exoskeleton device include a supporting portion, a cushion, a tying portion, a telescopic rod, and a bearing rod.

驱动器—串联弹性驱动器 SEA

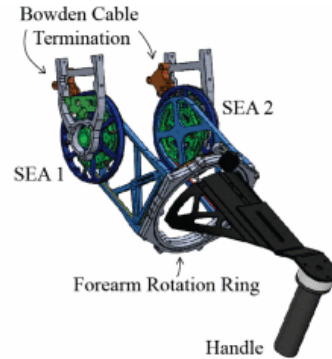
Design of upper limb rehab robotics

● Actuators—Series Elastic Actuator (SEA)

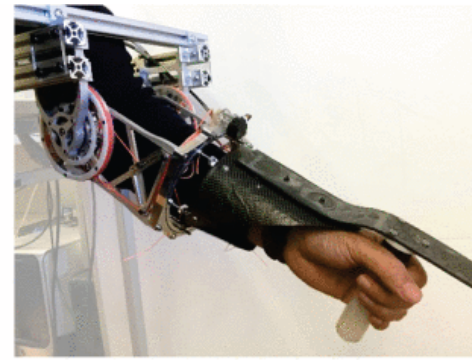
右图：一款肘外骨骼用SEA+绳索差动机理（TRO 2019论文）——两个SEA通过Bowden绳索驱动前臂旋转环。



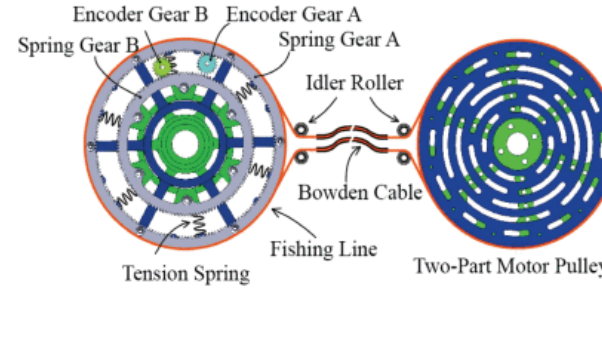
(a) Tethered system



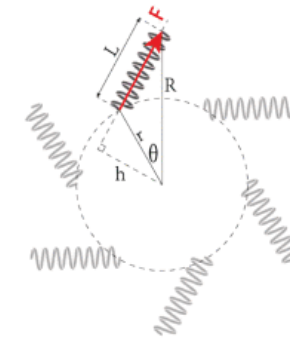
(b) CAD illustration



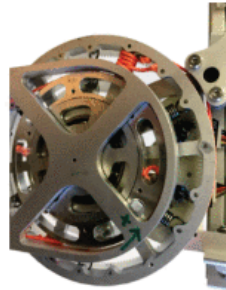
(c) Human arm in prototype



(a) Series elastic actuator schematics



(b) Geometry model for stiffness estimate



(c) Spring prototype

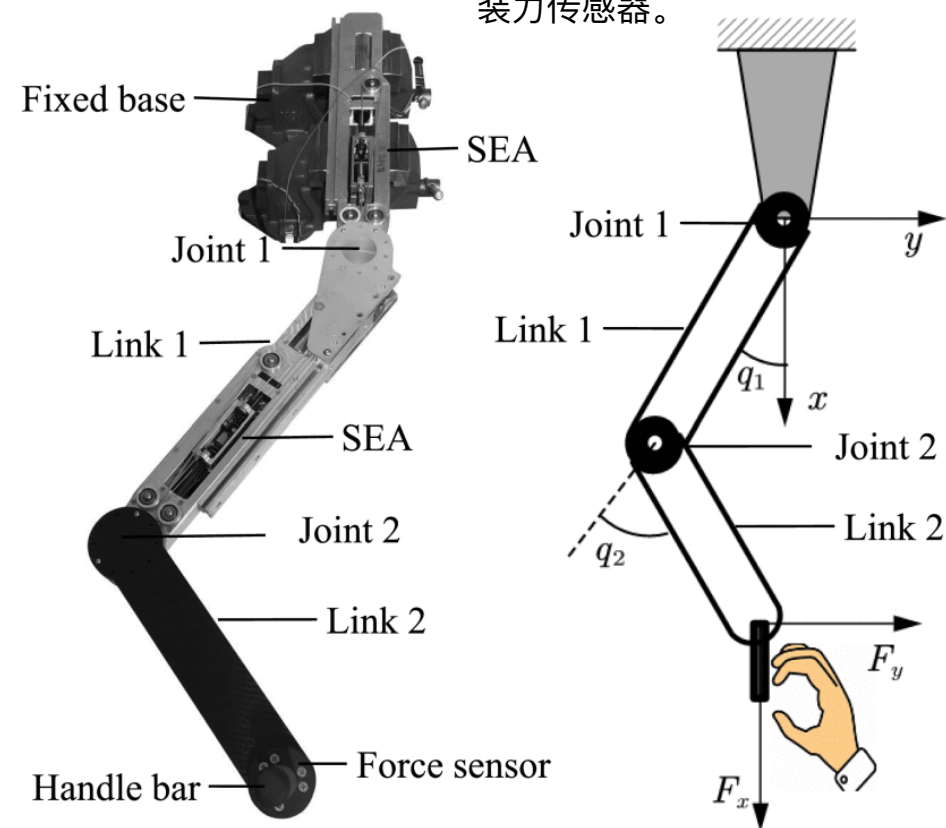
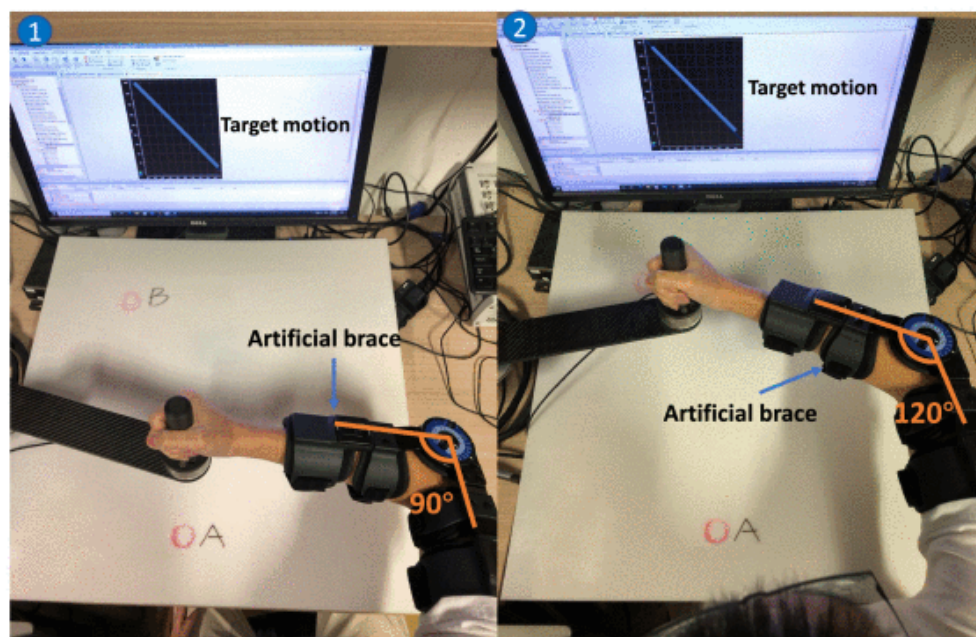
SEA: 电机+减速器后串联弹性元件（弹簧），提供柔顺交互界面。 [TRO, 2019]

- **SEA-driven:** SEA is a type of actuator that incorporates an **elastic element in series** with the motor and gearbox. The primary purpose of the elastic element, which is often a **spring**, is to provide a **compliant interface** with whatever the actuator is driving or manipulating.

- 新型SEA可高精度、高鲁棒性控制交互力矩
- SEA在物理人机交互（pHRI）中优势：高力/力矩保真度、高柔顺性、良好反向驱动性、抗冲击
- 这些特性带来更高的控制设计灵活性和人机安全性

Design of upper limb rehab robotics

● Case study—(SEA-driven)



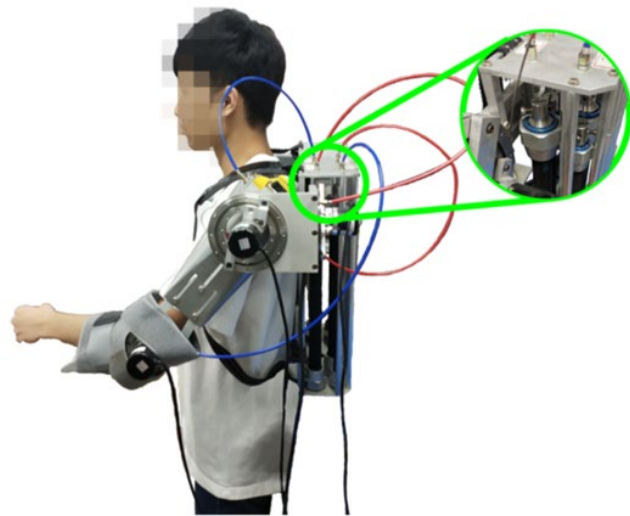
右图：两自由度平面外骨骼，关节1和关节2各装一个SEA，末端握把处装力传感器。

- **Case study:** a novel SEA can control the interaction torque with high accuracy and robustness.
- SEA has superior advantages in physical human–robot interaction (pHRI), such as high force/torque fidelity, high compliance, good backdrivability, and tolerance to shocks. These qualities bring in higher flexibility of control design and human–robot safety.

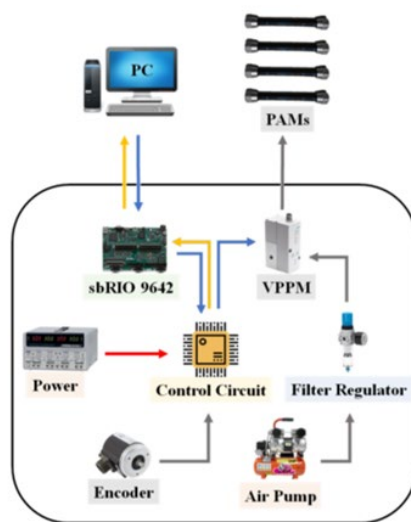
气动驱动器

Design of upper limb rehab robotics

● Actuators—Pneumatic Actuators

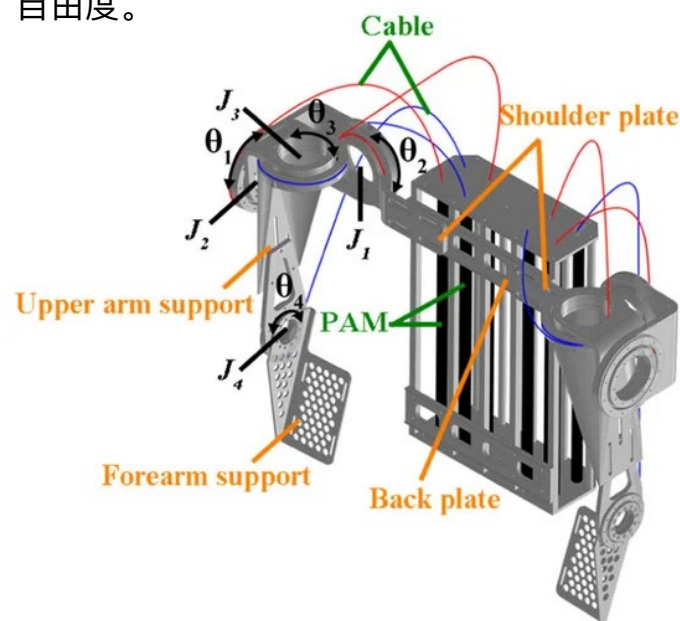


(a)



(b)

一款肩外骨骼用多根PAM（气动肌肉）通过绳索驱动肩关节多自由度。



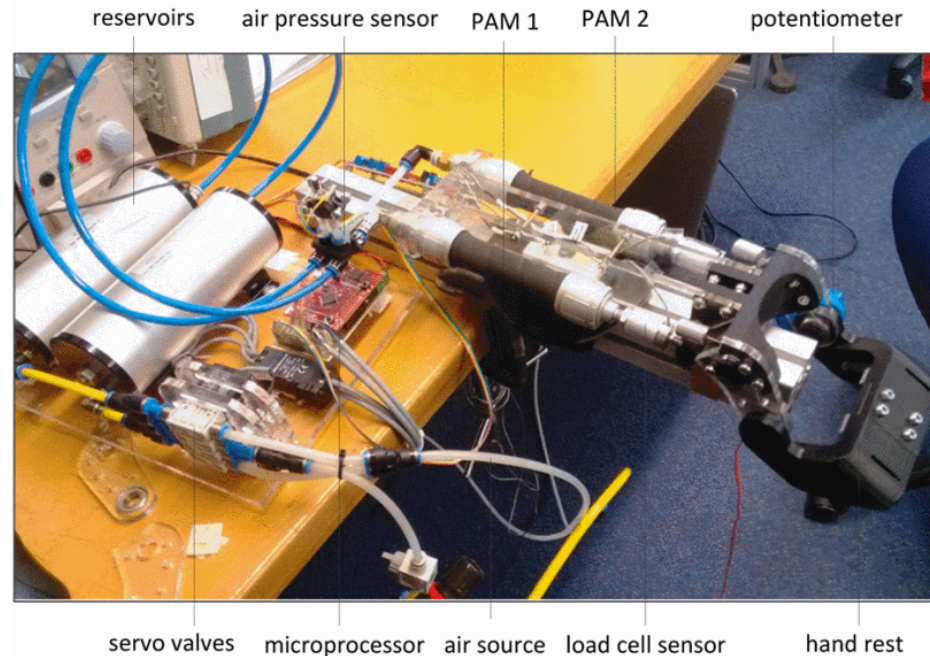
气动驱动器：气动肌肉执行器通过使用压缩气体使套管内的气囊膨胀，从而在径向扩张时实现轴向收缩，已成功应用于机器人及机器人康复治疗领域。

- **Pneumatic actuators:** Pneumatic muscle actuators inflate a sheathed bladder by employing a compressed gas to make it contract lengthwise when expanded radially and have been successfully applied in robotics and robotic therapy.

Design of upper limb rehab robotics

● Case study—(Pneumatic muscle actuators) 气动人工肌肉

气动人工肌肉：压缩气体充入套管气囊，径向膨胀同时轴向收缩，模拟人体肌肉收缩。



两根Festo Fluidic Muscle作为拮抗驱动器，固定在患者前臂背面塑料板上，通过钢丝绳连接机械铰链。一根PAM收缩时顺时针转铰链，另一根逆时针转。

组件：储气罐、气压传感器、PAM1/PAM2、电位计、伺服阀、微处理器、气泵、拉压力传感器、手托。

Two Festo Fluidic Muscles are used as the antagonistic actuators and are placed on a plastic plate, which is strapped on the back of the patient's forearm. Both muscles are attached to a mechanical hinge through a steel wire. The wires are guided around the cylindrical hinge in a way, so that one PAM rotates it in a clockwise direction, and the other PAM in a counter clockwise direction when contracting.

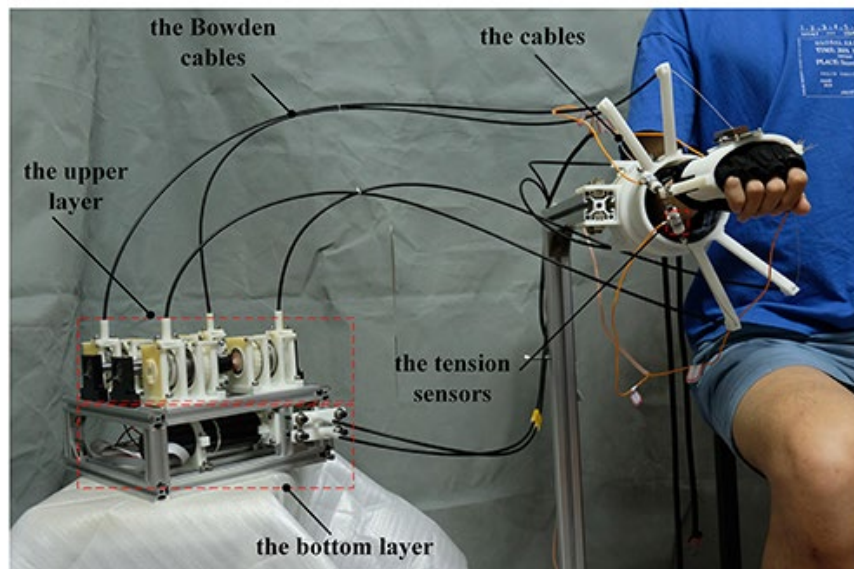
绳索驱动

Design of upper limb rehab robotics

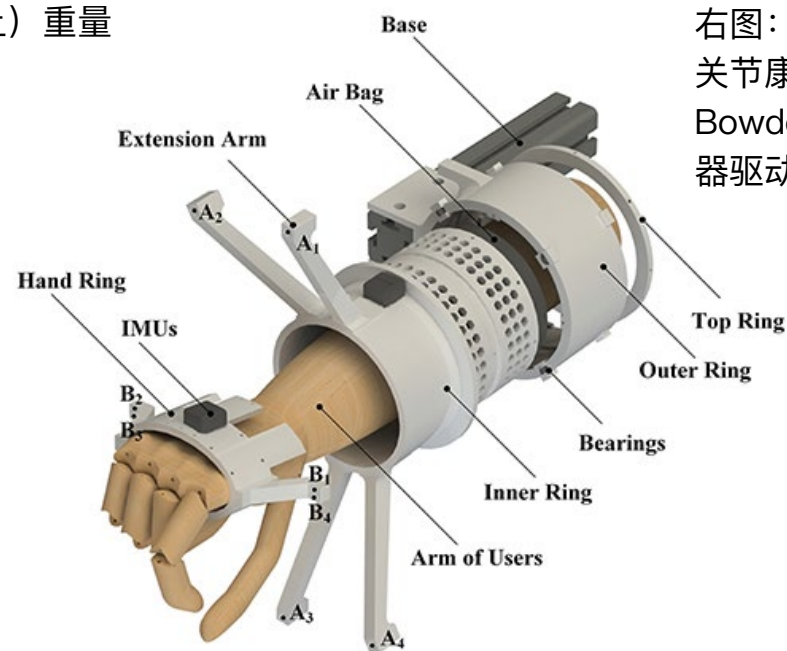
● Actuators— Cable-driven Actuators

- 绳索通过穿戴衣物上的锚点施加可控力
- 电机布置在近端，通过柔性绳索传力到关节，减少远端（手臂上）重量

A



B



右图：一款三自由度腕关节康复外骨骼，用Bowden绳索+张力传感器驱动。

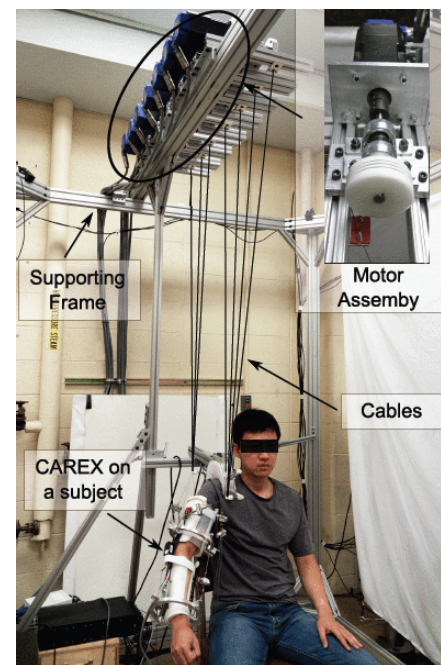
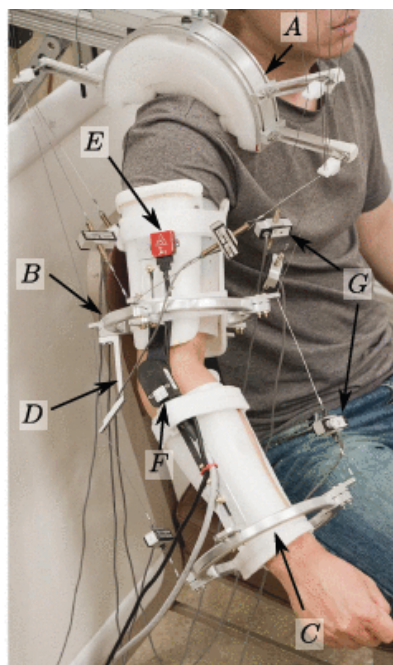
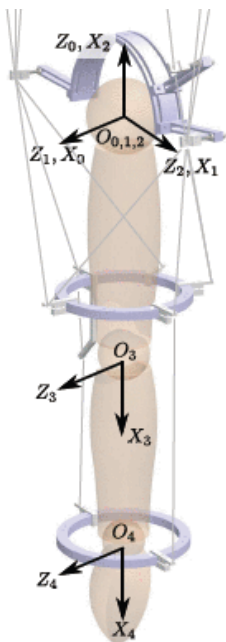
- **Cable-driven:** can apply controlled forces to the wearer by attaching at anchoring points in the wearable garment.
- Cable minimize the distal mass that is attached to the wearer through more proximally mounted actuation systems and flexible transmissions that transmit power to the joints.

实例 CAREX

Design of upper limb rehab robotics

● Case study— (Cable-driven rehab robotics)

CAREX是伯克利的绳索驱动外骨骼代表——所有电机都在天花板/支架上，手臂上只有轻的绑带和绳索。



刚性固定在臂矫形器上，紧绑在人手臂上。绳索路由点针对ADL典型手臂"张紧"工作空间优化。电机组悬挂在支架上方，不戴在 person 上。

- CAREX was rigidly attached to an arm orthosis which can be tightly strapped on the human arm. The cable routing points were optimized for large "tensioned" workspace of the arm for typical Activities of Daily Living (ADLs).

VR辅助治疗

Design of upper limb rehab robotics

- VR (Assisted therapy device)



Rehab Therapy is so boring! Can anyone help?

Virtual-Reality(VR) can help!



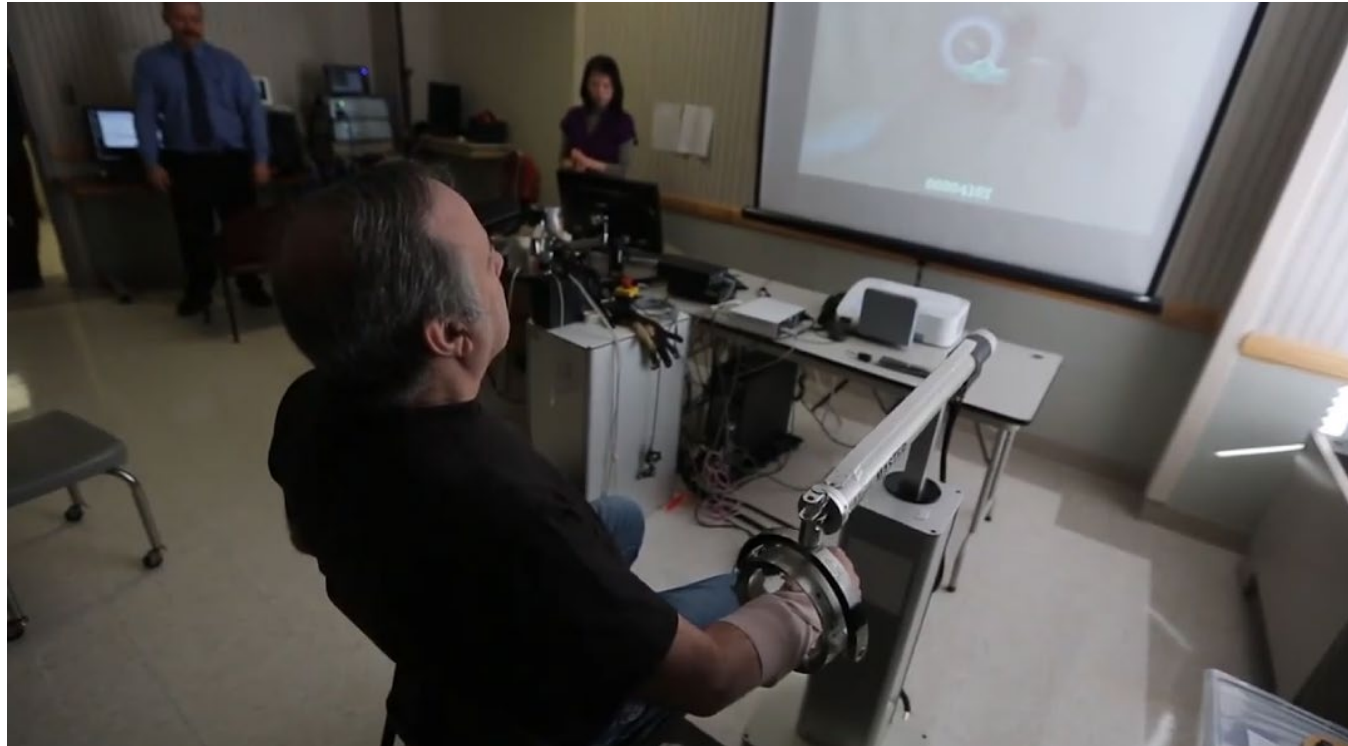
VR: 沉浸式多感官方式，为开发更经济、定制化、个性化的上肢居家治疗方案提供研究方向

- **VR:** an immersive, multi-sensory approach and provides future avenues for research in developing more cost-effective, tailored, and personalized upper limb solutions for home therapy applications.

AI应用

AI in upper limb rehab robotics

● VR/AR

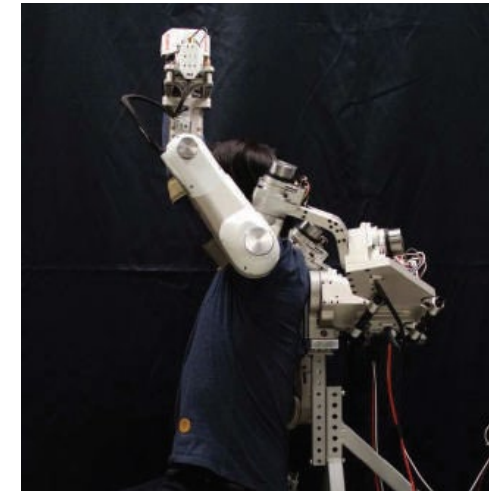
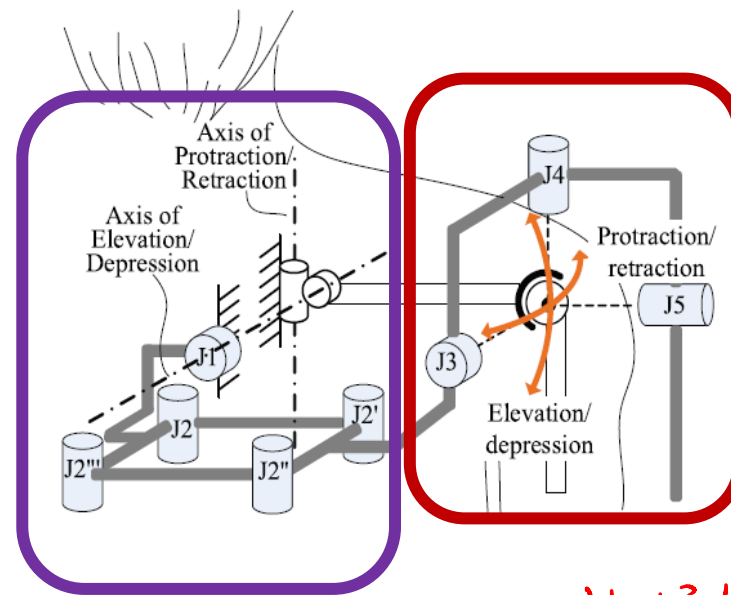
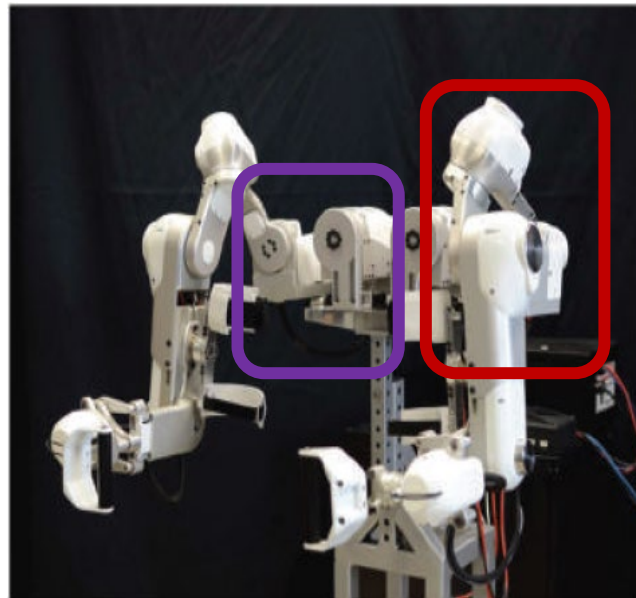


• VR/AR: 一种沉浸式、多感官的解决方案，为开发更具成本效益、个性化和定制化的上肢康复治疗方案提供了未来研究方向。

- **VR/AR:** an immersive, multi-sensory approach and provides future avenues for research in developing more cost-effective, tailored, and personalized upper limb solutions for home therapy applications.

具有明确功能的 上肢辅助机器人 Targeted function of upper limb rehab robotics

- Exoskeleton-based robot for shoulder 肩部靶向外骨骼



Harmony (Kim and Deshpande, 2017)

被动量力补偿，防脱位。

- **Harmony**: active and passive assistance; achieve **Scapula** & **Shoulder** movements; Note that shoulder misalignment may occur if scapula movements are not achieved.

- 主动+被动辅助
- 实现肩胛骨+肩关节运动
- 注意：如果不实现肩胛骨运动，会发生肩关节错位

Adapted from B. Kim and A. D. Deshpande, "An upper-body rehabilitation exoskeleton Harmony with an anatomical shoulder mechanism: Design, modeling, control, and performance evaluation," *The International Journal of Robotics Research*, vol. 36, no. 4, pp. 414-435, 2017.

Shoulder rehab exoskeleton: case study

Harmony是少数专门驱动肩胛骨的外骨骼——紫色框是肩胛骨的上提/下降、前伸/后缩自由度；红色框是盂肱关节。这是解决肩关节错位问题的经典方案。



通过重复性、任务特异性运动，外骨骼促进神经可塑性，帮助大脑在损伤后重新组织和适应。

(ETH ANYexo)

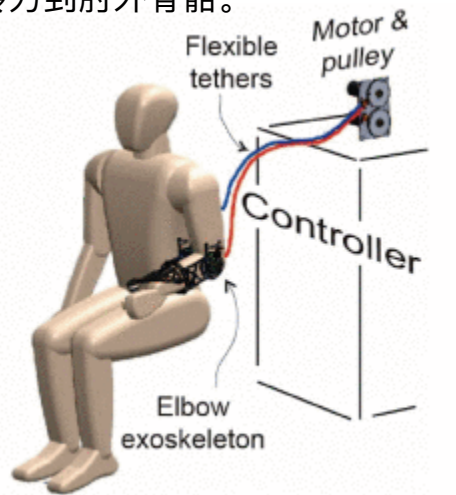
- By providing repetitive and task-specific movements, exoskeletons contribute to neuroplasticity, facilitating the brain's ability to reorganize and adapt after injury.

肘部朝向骨骼。

Targeted function of upper limb rehab robotics

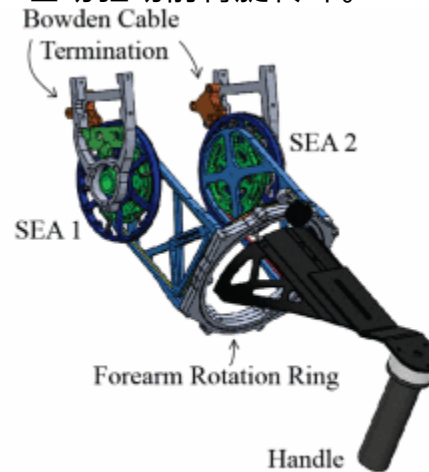
● Exoskeleton-based robot for elbow

左图：tethered系统，电机在远端通过柔性系绳传力到肘外骨骼。

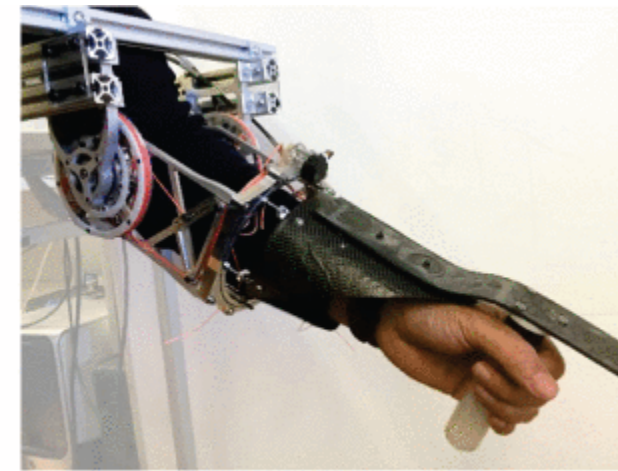


(a) Tethered system

两个SEA通过Bowden绳索差动驱动前臂旋转环。



(b) CAD illustration



(c) Human arm in prototype

小臂+手有不小自重。

- 主动+被动辅助
- 实现肘屈伸+前臂旋前旋后
- 注意避免肘关节错位

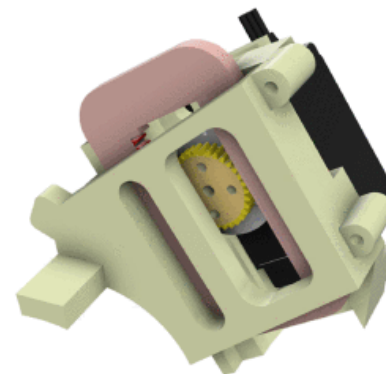
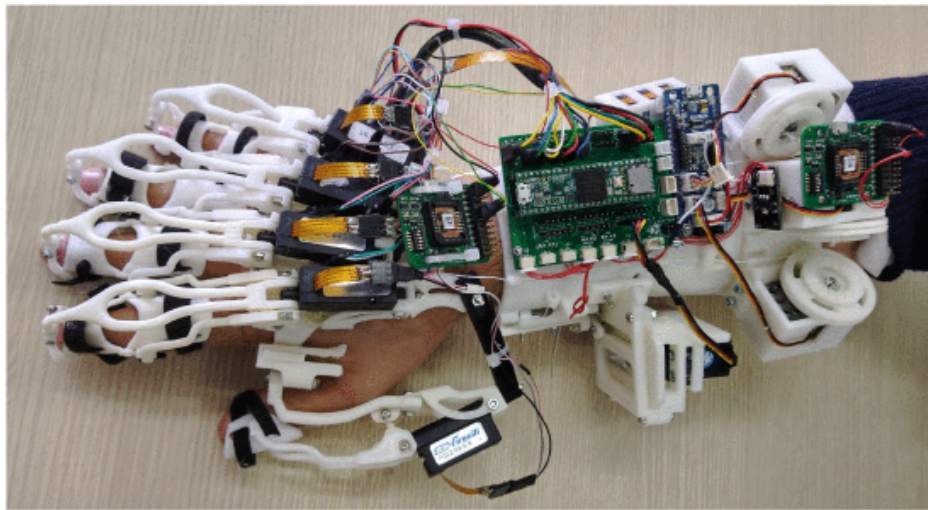
Elbow Rehab Robot (Chen et al., 2019)

- **Elbow Rehab Robot:** active and passive assistance; achieve flexion/extension and forearm supination/pronation
Avoid misalignment which may occur to elbow.

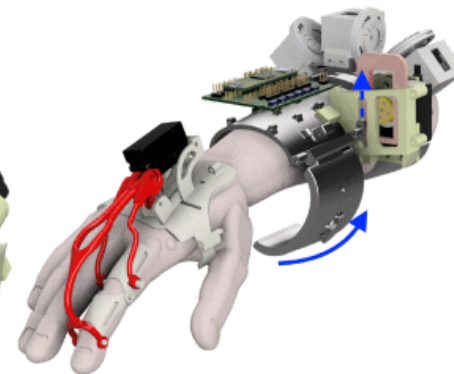
腕部靶向外骨骼。

Targeted function of upper limb rehab robotics

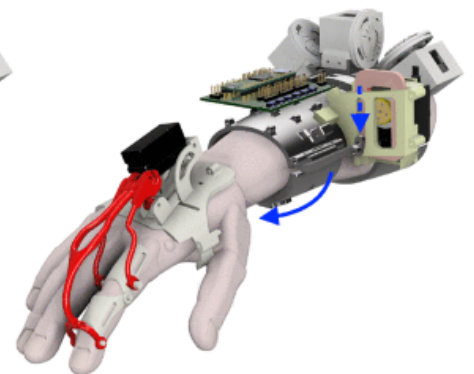
- Exoskeleton-based robot for wrist



(a)



(b)



(c)

- 主动辅助
- 实现腕+前臂运动

Wrist Rehab Robot (Dragusanu et al., 2022)

- **Wrist Rehab Robot:** active assistance; achieve wrist & forearm movements

手指靶向外骨骼 — 软体手套 Targeted function of upper limb rehab robotics

● Exoskeleton-based robot for finger

MCP: 掌指关节

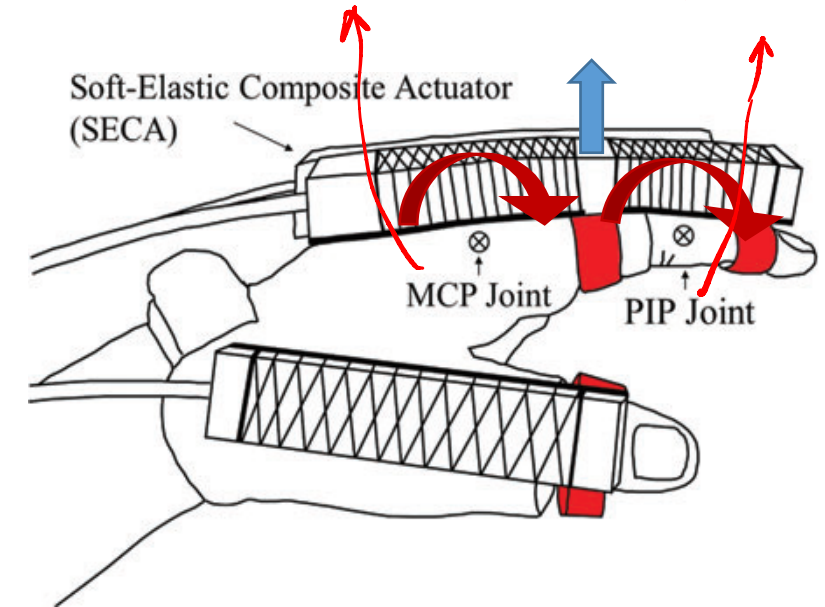
PIP: 近端指
尖关节。

软体复合驱动器

Soft-Elastic Composite Actuators
(SECA)

Fingerless glove

Tubing
气管

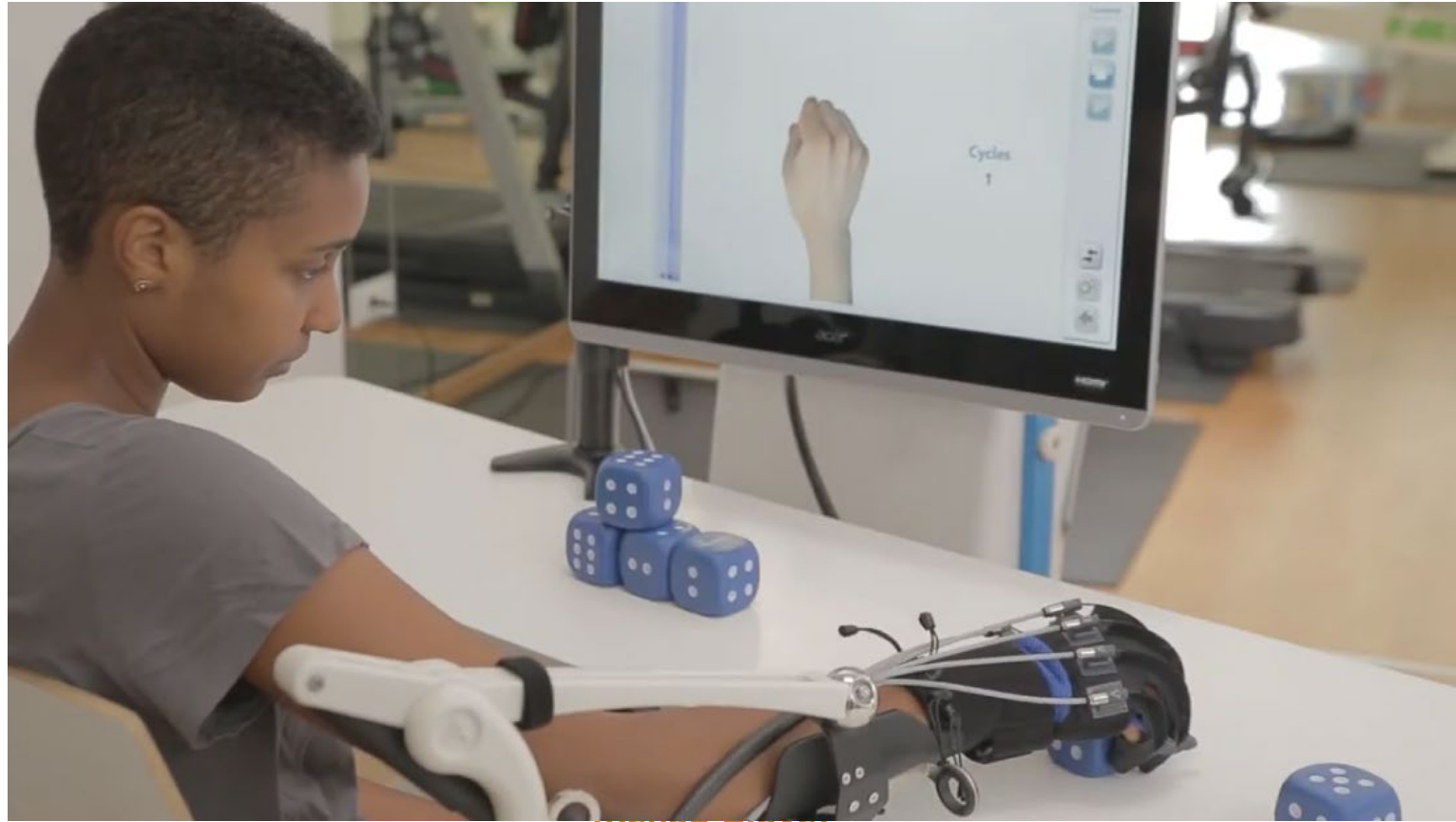


- 主动+被动辅助• 实现手指运动
- SECA (Soft-Elastic Composite Actuator, 软弹性复合驱动器): 通入压缩空气时膨胀并弯曲, 驱动MCP和PIP关节
- 戴无指手套上

- **Finger Rehab Robot:** active & passive assistance; achieve finger movements; SECA expands and bends when applying compressed air

案例

Hand rehab robots: case study



患者戴手部外骨骼做桌上骰子任务，屏幕显示虚拟手。

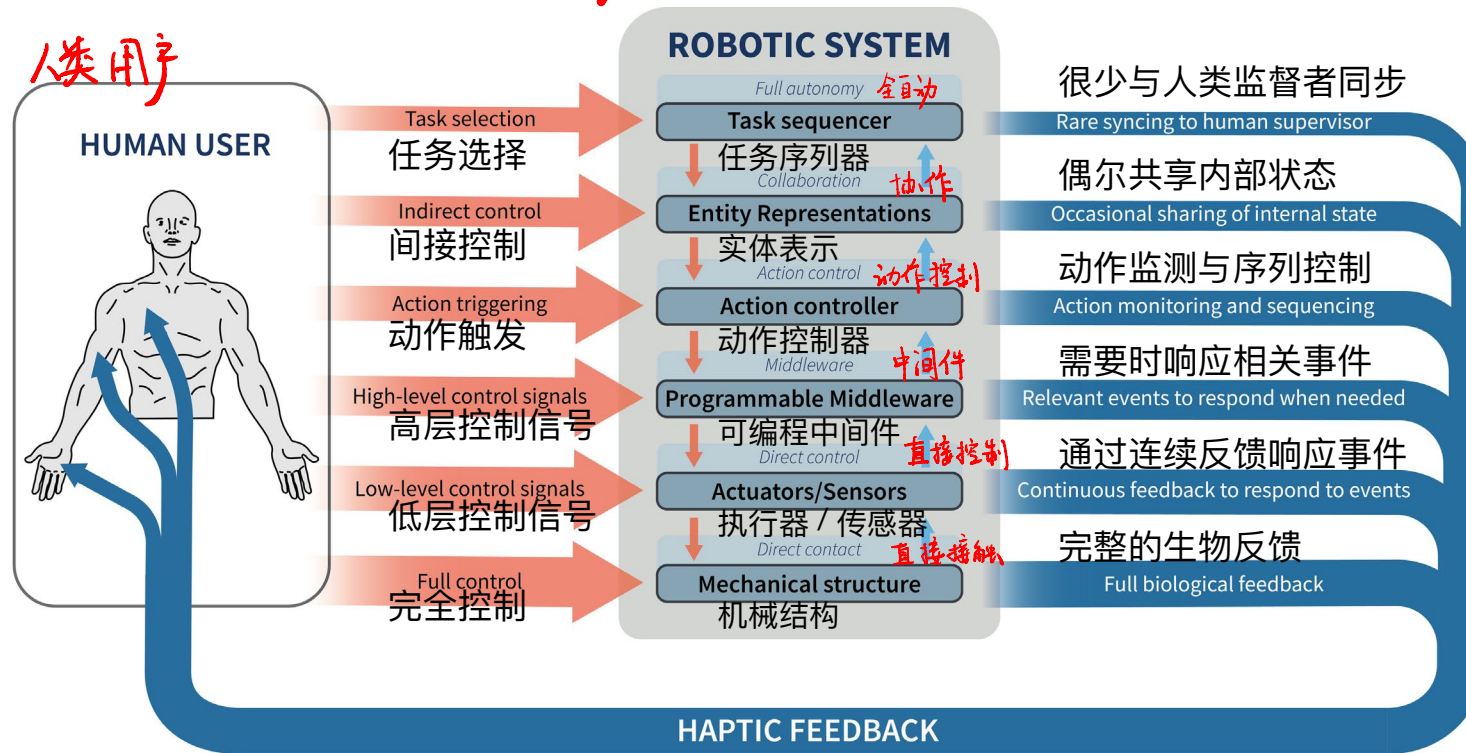
任务导向训练可根据患者情况定制：功能治疗可以是主动辅助、被动或双侧。

Task oriented exercises can be customized depending on the condition of the patients: functional therapy can be active-assisted, passive or bilateral.

上肢辅助机器人的控制

Control of upper limb rehab robotics

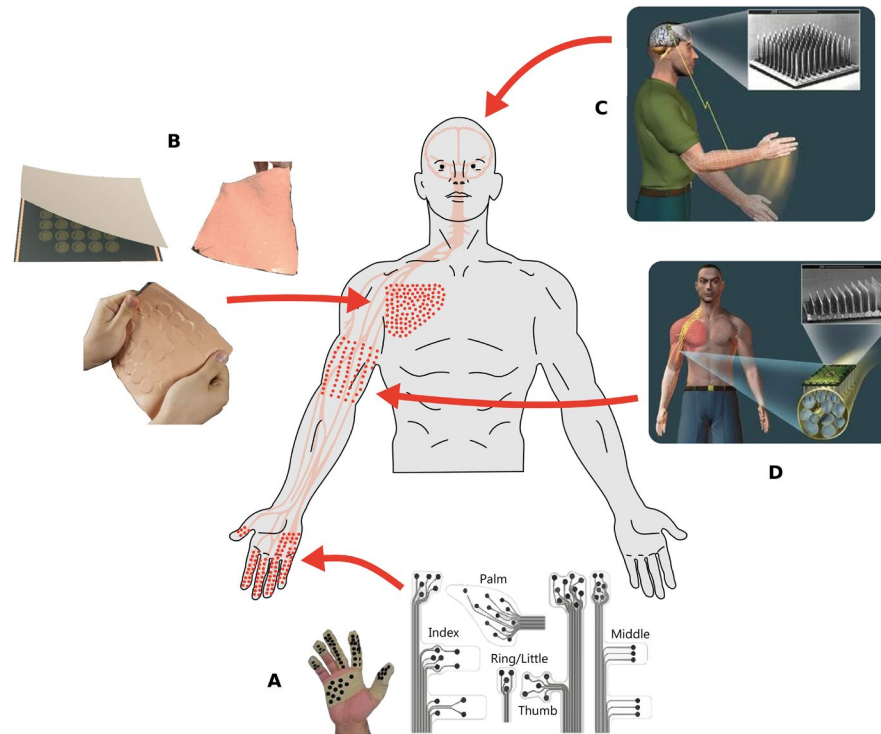
● Hierarchical Controller 多层控制器



- **Hierarchical controller:** consists of high-level (for human motion intention recognition), middle-level (for action control), and low-level control (for control commands execution) • 高层：人体运动意图识别。 • 中层：动作控制。 • 低层：控制命令执行

Control of upper limb rehab robotics

- High-level control (human motion intention recognition)



高层控制主要由人体运动意图检测传感器组成：触觉传感器、IMU、深度传感器、EMG肌电、EEG脑电。

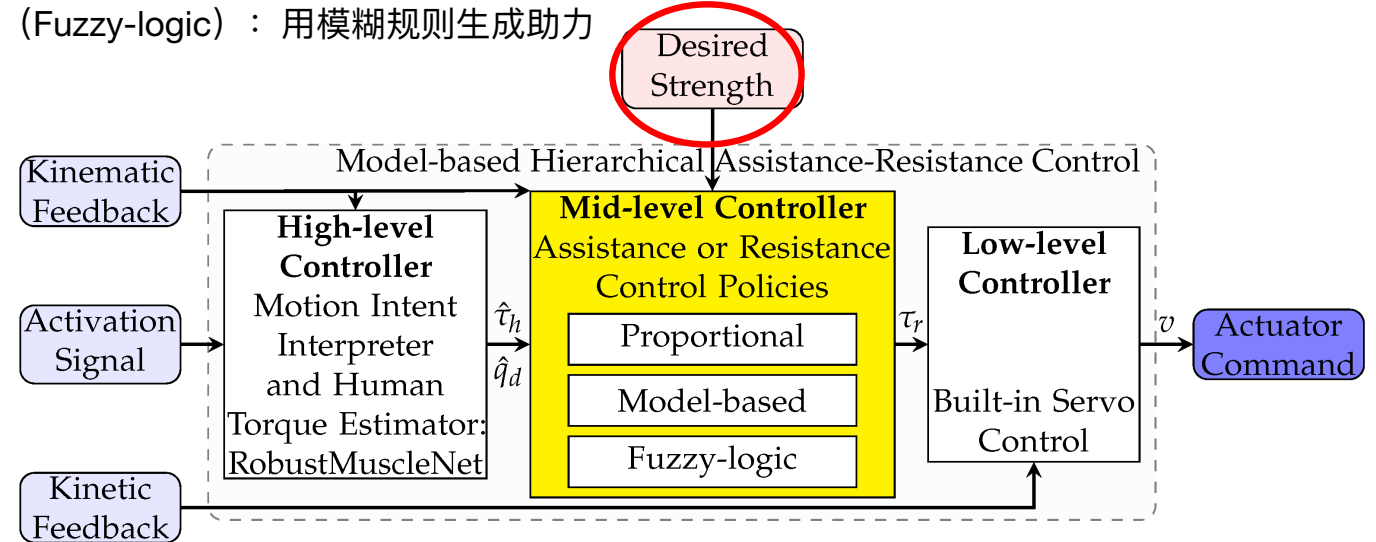
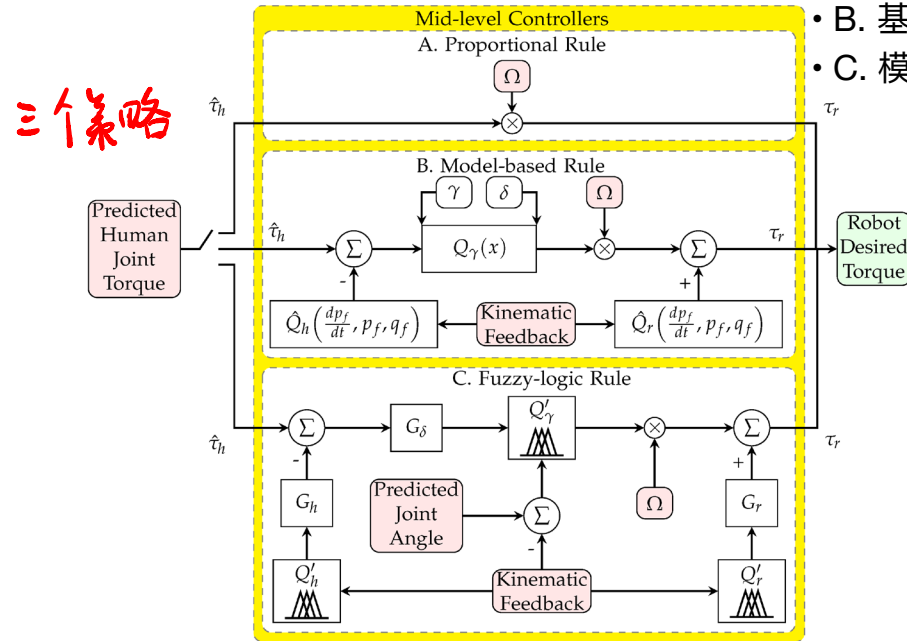
- High-level control: mainly consisting of sensor for human motion intention detection, such as tactile sensor, IMUs, depth-sensors, EMG, EEG

Control of upper limb rehab robotics

● Mid-level control (action control of upper limb)

三种中层助力控制策略：

- A. 比例规则 (Proportional) : 按比例输出助力力矩
- B. 基于模型 (Model-based) : 用肌肉骨骼模型预测关节力矩
- C. 模糊逻辑 (Fuzzy-logic) : 用模糊规则生成助力



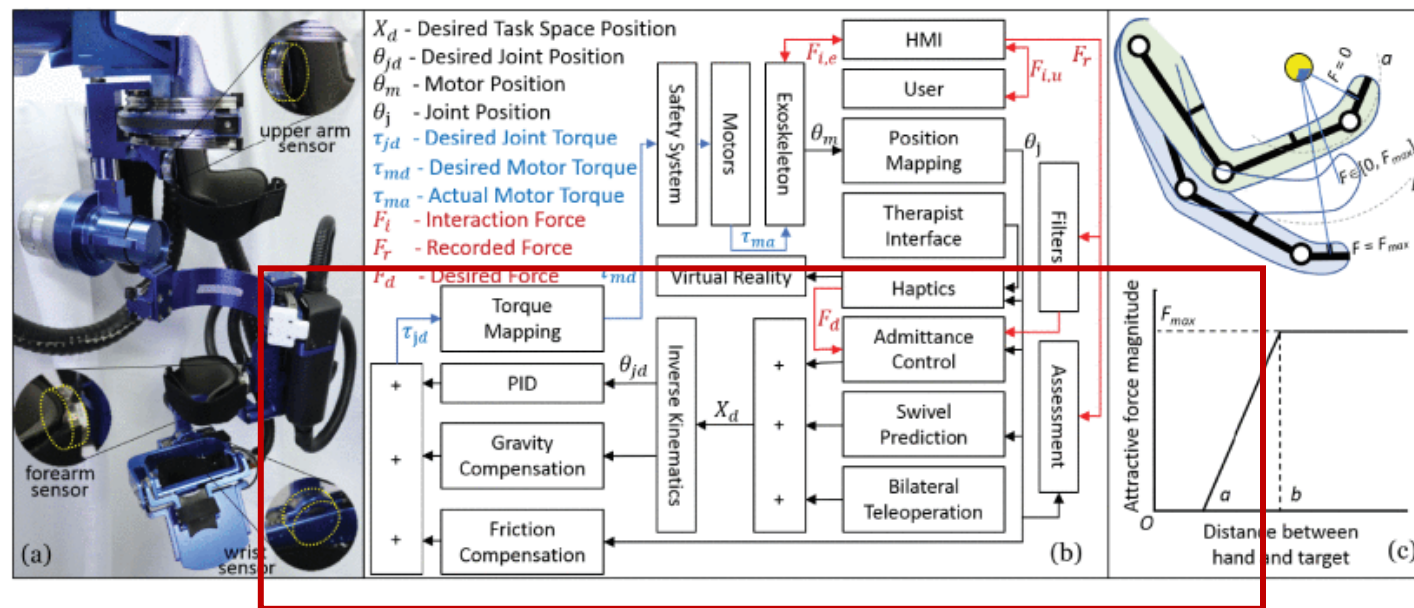
中层控制器为低层控制器生成助力力矩指令。

- **Motion control:** some mid-level controllers are designed for assistive torque generation for low-level controllers.

低层控制 — 力矩控制.

Control of upper limb rehab robotics

● Low-level control (torque control)



PID力矩映射+重力补偿+摩擦补偿→逆运动学→
导纳控制+滑动预测+双边遥操作→外骨骼关节。

力矩控制：用控制算法和机制调节外骨骼各关节输出的力矩。

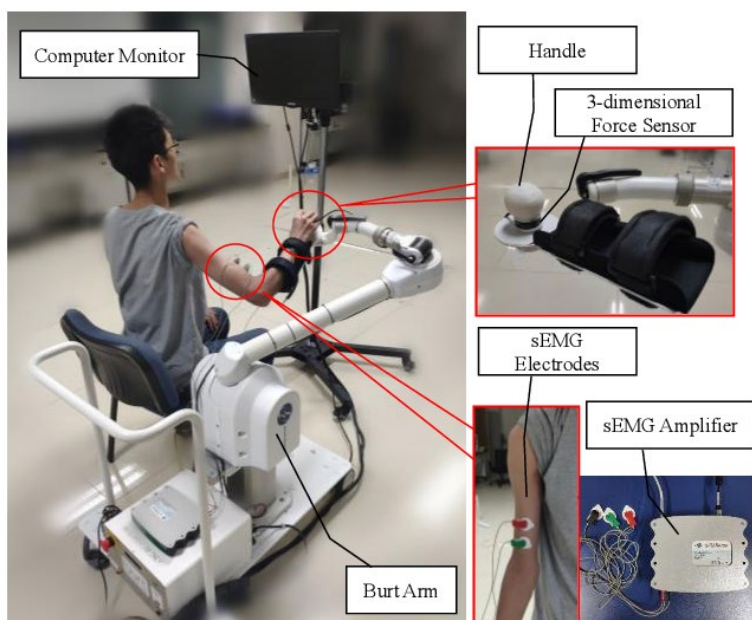
- **Torque control:** refers to the use of control algorithms and mechanisms that regulate the torque applied at the joints of the exoskeleton.

EMG 驱动控制。

Control of upper limb rehab robotics

● EMG-driven control

BurtArm外骨骼+sEMG电极+三维力传感器。

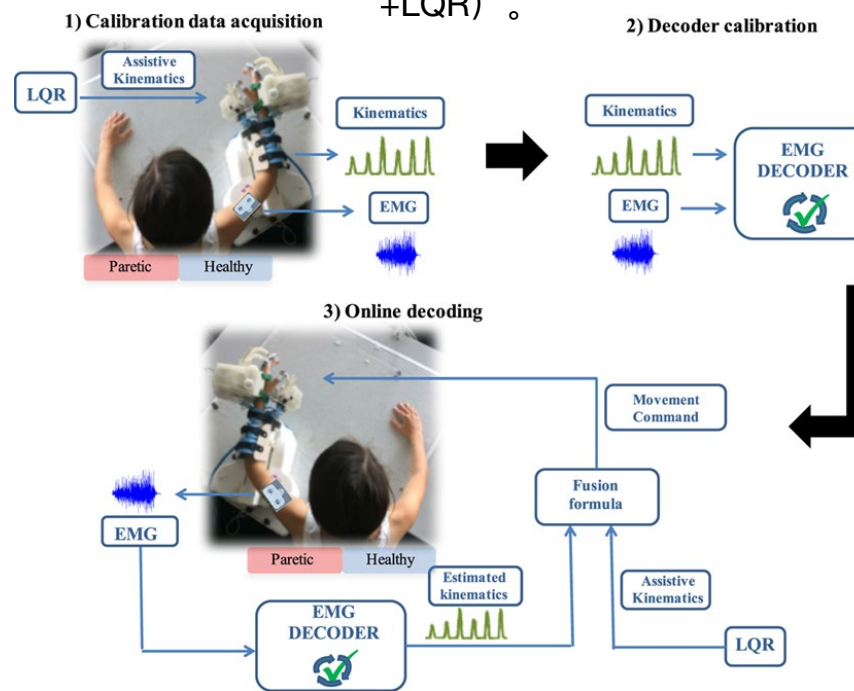


肌电信号能解读更复杂的肌肉模式来判断用户意图，实现更直观的康复机器人控制。

sEMG-driven control

- **EMG-driven control:** Myoelectrical signal is capable of interpreting more complex muscle patterns to determine the user's intent and leads to a more intuitive control of the rehab robotics.

右图流程：1) 校准数据采集（患侧+健侧）→ 2) 解码器校准 → 3) 在线解码（融合公式+LQR）。



Control of upper limb rehab robotics

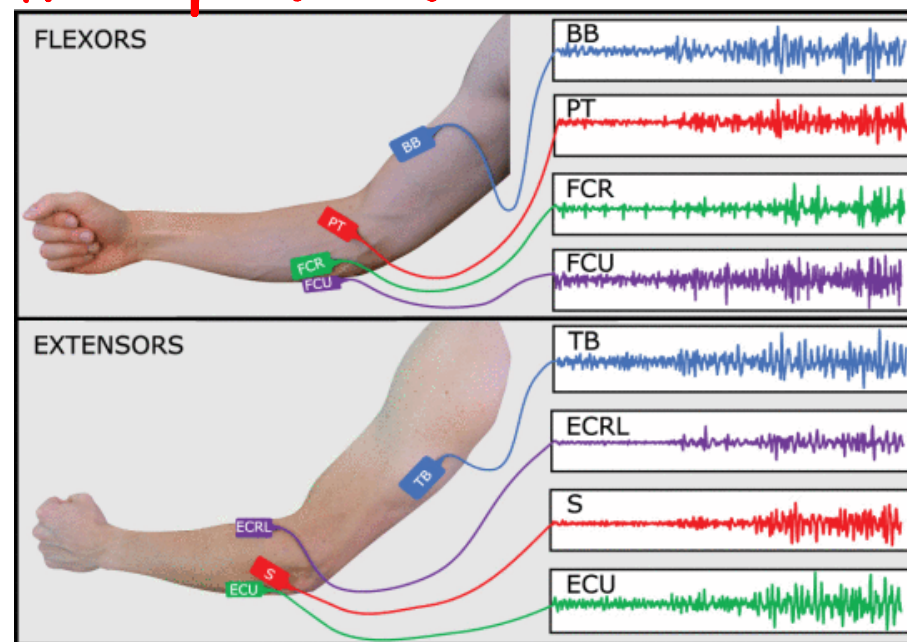
● EMG-driven control



- 肌电接口：肌电接口能够识别更复杂的肌肉模式，以确定用户的意图，从而实现对康复机器人更直观的控制。

[TNSRE, 2020]

肌电信号图



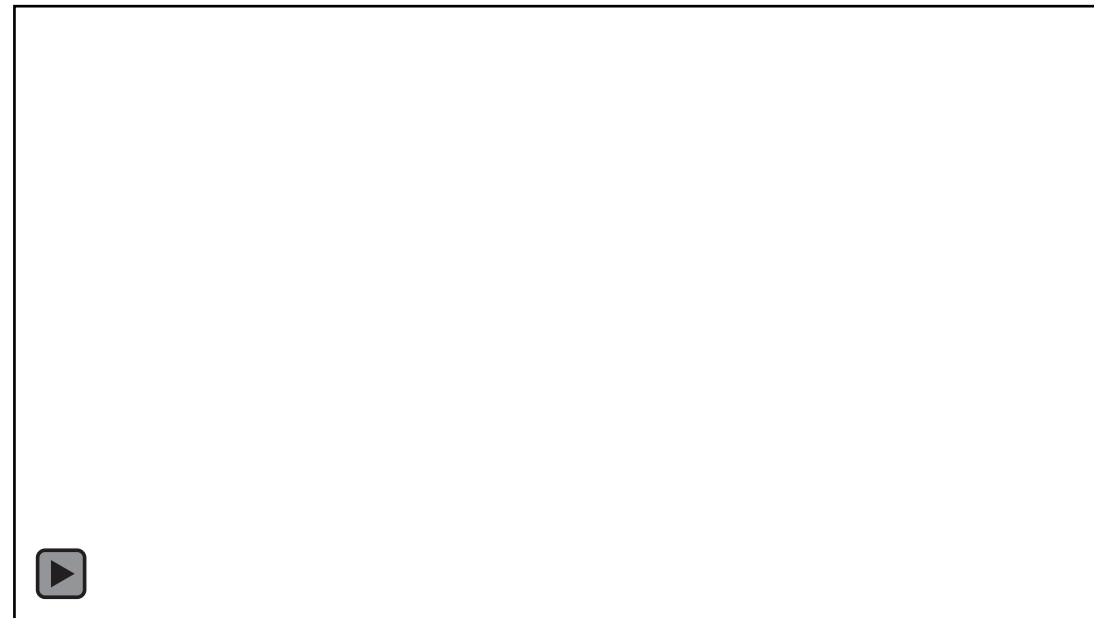
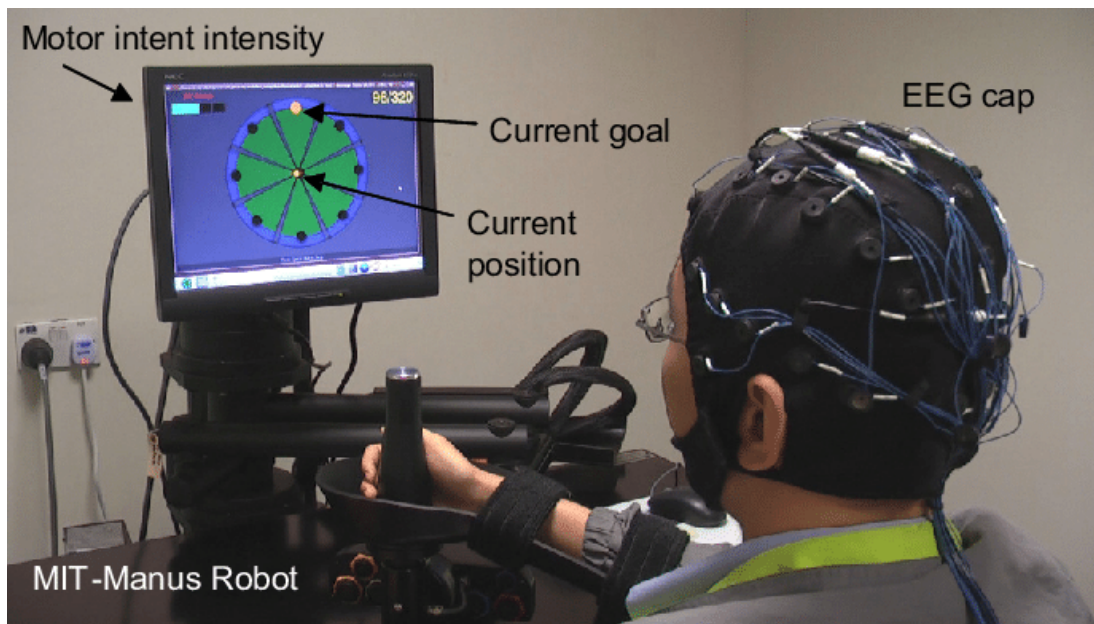
前臂贴上多通道EMG电极，屈肌群（BB肱二头肌、PT肱桡肌、FCR桡侧腕屈肌、FCU尺侧腕屈肌）和伸肌群（TB肱三头肌、ECRL桡侧腕长伸肌、S、ECU尺侧腕伸肌）的肌电波形。

- **Myoelectrical-interface:** Myoelectrical-interface is capable of interpreting more complex muscle patterns to determine the user's intent and leads to a more intuitive control of the rehab robotics.

脑机接口

Control of upper limb rehab robotics

● Brain-computer-interface (BCI)



(MIT-Manus Robot) 患者戴EEG脑电帽，操作MIT-Manus机器人，屏幕显示环形光标任务——大脑想往哪动，光标就往哪走，机器人辅助手臂完成。 (Rehab training based on feedback of BCI)

- **Brain-computer-interface:** use of **input neural signals** for the generated output movement.
- **BCI** involves signal processing, intent interpretation, training, adaptation and learning. Various AI technologies exist in these process.

- BCI: 用输入神经信号生成输出运动
- 涉及信号处理、意图解读、训练、自适应和学习，其中用到各种AI技术

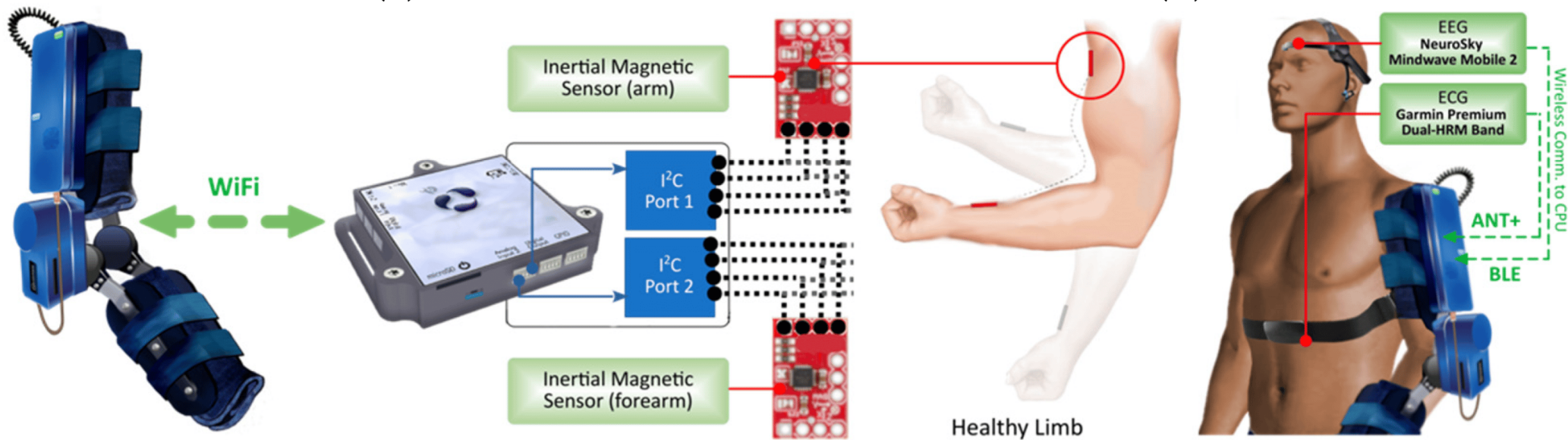
传感器融合.

Control of upper limb rehab robotics

● Sensor Fusion

图：臂+前臂各贴一个惯性磁传感器（通过I2C连主控），WiFi无线传输；健侧肢体也戴

(a) 传感器，同时EEG脑电帽+ECG心率带无线通信到CPU。(b)



[Sensors, 2021]

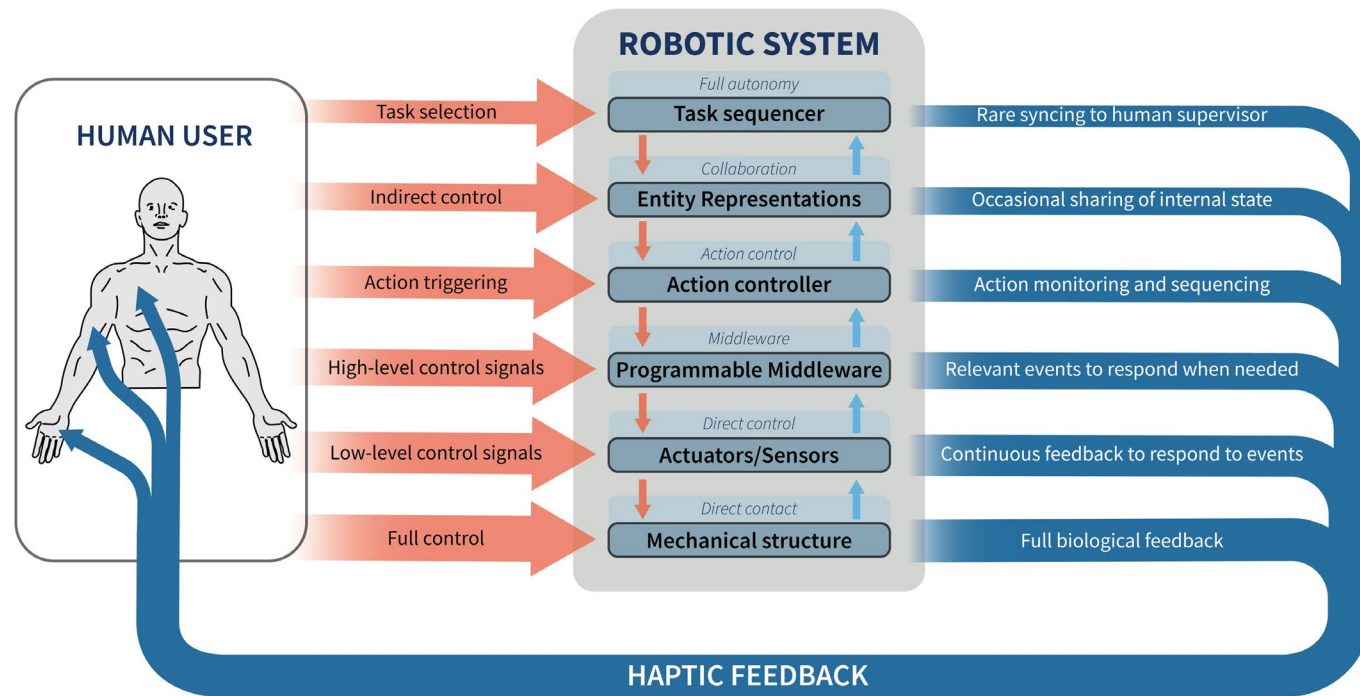
AI融合多类传感器数据（IMU、力传感器、EMG、EEG、相机），更全面理解用户运动和机器人空间位置。

- **Sensor Fusion:** AI can integrate data from multiple types of sensors (like IMU, force sensors, EMG, EEG, and cameras) to create a more holistic understanding of the user's movement and the robot's position in space.

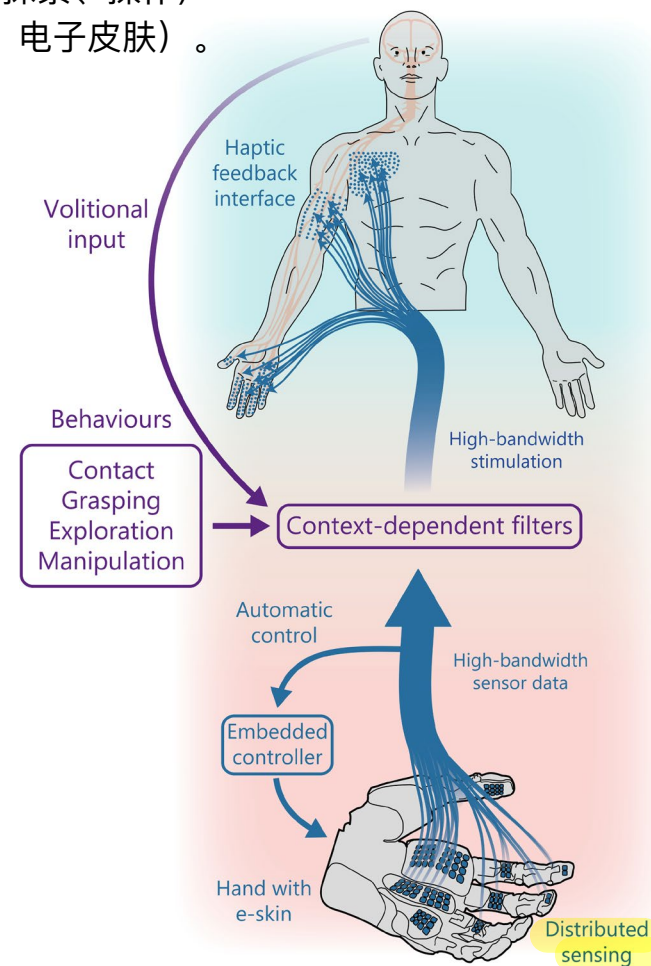
控制架构：人在回路。

Control of upper limb rehab robotics

● Control framework: Human-in-the-loop



右图：有意识层面（自主输入、触觉反馈、高带宽刺激）→
上下文相关过滤器 → 行为（接触、抓握、探索、操作）→
无意识层面（嵌入式控制器、分布式传感、电子皮肤）。



人在回路：外骨骼动作实时受用户意图和生理信号影响引导。人机主动参与协作，形成闭环系统，持续适应用户运动、意图和需求。

Subconscious level

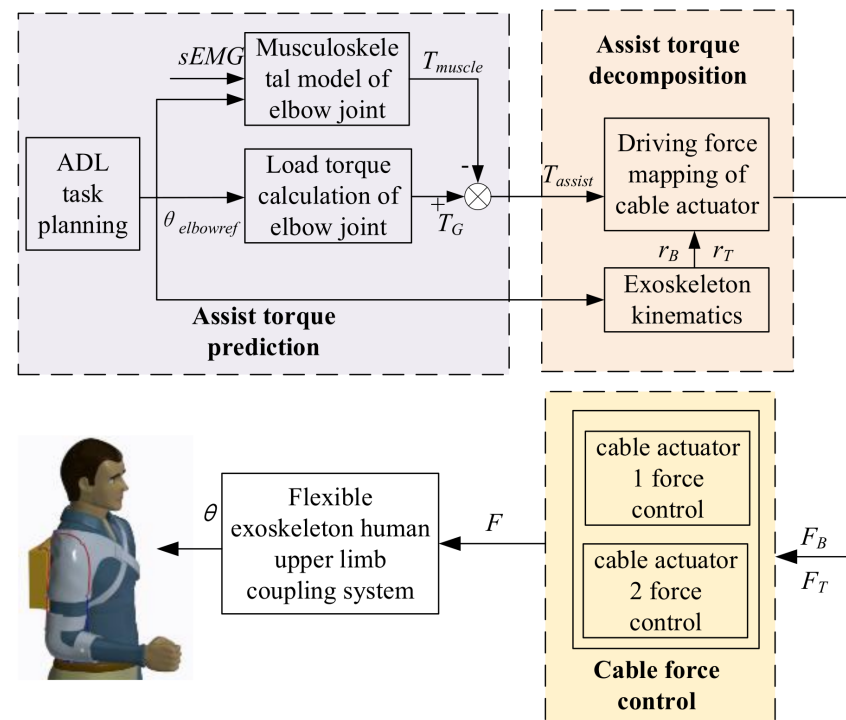
- **Human-in-the-loop:** refers to a control strategy where the exoskeleton's actions are influenced or guided by the user's intent and physiological signals in real-time. This approach involves active participation and collaboration between the exoskeleton's control system and the user, creating a closed-loop system that continuously adapts to the user's movements, intentions, and needs.

按需辅助 (AAN)

Control of upper limb rehab robotics

● Control framework: Assist-as-needed

框图：sEMG+ADL任务规划
→肘关节肌肉骨骼模型+负载力矩计算→助力力矩预测
→绳索驱动器驱动力映射→柔性外骨骼人机耦合系统。



AAN是康复机器人控制的核心理念——患者自己能动的部分机器人不帮，快动不了了才帮一把。目的是逼患者主动参与，而不是机器人全带动。

- **Assist-as-needed:** a concept commonly used in upper-limb rehabilitation robotics to tailor assistance to the user's needs during the rehabilitation process. The goal is to provide support and guidance to the user, promoting active participation while ensuring safety and effectiveness. 按需辅助 (AAN)：上肢康复机器人常用概念，根据用户康复过程中的需求定制助力。目标是提供支撑和引导，促进主动参与，同时

确保安全有效。

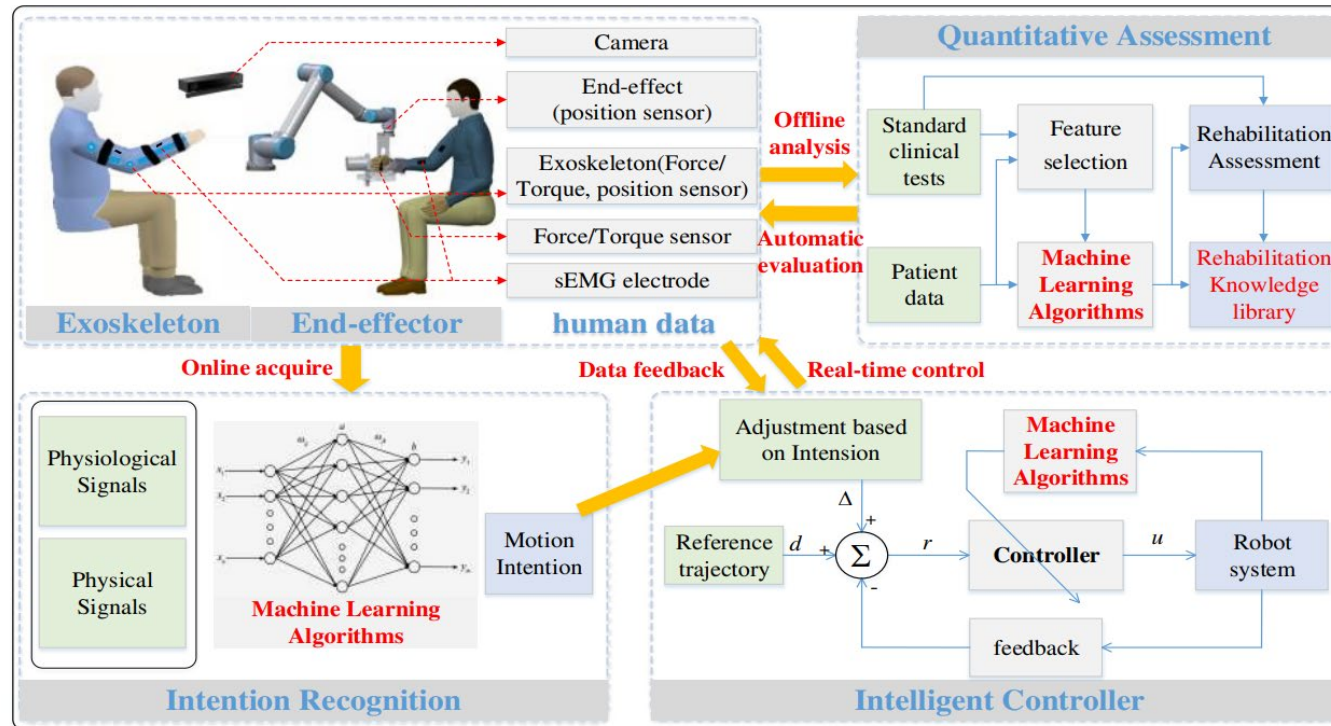
Hu B, Zhang F, Lu H, et al. Design and assist-as-needed control of flexible elbow exoskeleton actuated by nonlinear series elastic cable driven mechanism[C] Actuators. MDPI, 2021, 10(11): 290.

AI - 机器学习

AI in upper limb rehab robotics

● Machine Learning

上方：外骨骼+末端执行器采集人数据（相机、末端位置、力/力矩传感器、sEMG）→ 离线分析（标准临床测试、特征选择、机器学习算法→ 康复评估



下方：在线采集生理信号+物理信号→机器学习算法→运动意图识别→智能控制器（参考轨迹+力矩调整）→机器人系统实时反馈

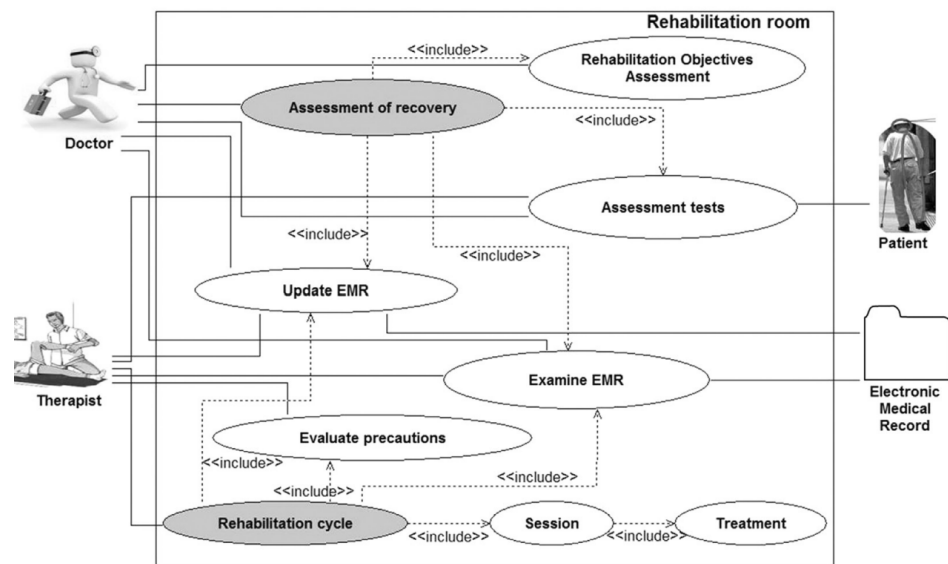
- **Machine learning:** Machine learning technologies such as classification, regression, and cluster are widely used in vision, intention recognition, EMG and EEG signal processing. 机器学习（分类、回归、聚类）广泛用于视觉、意图识别、EMG和EEG信号处理。

AI in upper limb rehab robotics

● GPT



右图：UML统一建模语言——康复室里医生、治疗师、患者、电子病历之间的流程交互图。



[Unified Modeling Language (UML)]

- GraspGPT提示LLM获取关于新动作概念的自然语言描述
- 语言描述将新概念与训练中已学概念连接，实现从已知概念到新概念的任务导向治疗泛化

- **GPT used in training:** (a) GraspGPT prompts an LLM to acquire language descriptions about the novel concept(s) in a natural language instruction given by the user. (b) Language descriptions connect the novel concept to its related concepts described during training, enabling the generalization of task-oriented therapy from known concepts to novel concepts.

挑战 1 - 成本

Challenges and future trends

● Challenge 1: Design (how to reduce cost)

三台上肢康复机器人价格：

- Reogo上肢康复机器人：1万美元
- ArmeoSpring：4万美元
- InMotion ARM：8万美元



(Reogo upper limb rehab robot)
10000 \$



(ArmeoSpring rehab robot)
40000 \$



(Imotion ARm rehab robot)
80000 \$

- **Design: high cost;** 高成本：电机、传动、传感器价格高，限制了动力外骨骼商业化普及。
- High price of the motor, transmission, and sensors limit the commercialization of the powered exoskeletons

挑战2 - 临床试验

Challenges and future trends

● Challenge 2: Clinical trials

上肢康复外骨骼需要大量受试者和临床试验来评估疗效。这是漫长而困难的过程。



The patient is diagnosed with sequelae of cerebral hemorrhage

图：Yeecon A6机器人在医院临床使用，患者是脑出血后遗症患者。

- Upper-limb rehabilitation exoskeletons require a large number of subjects and clinical trials to evaluate the performance. This is a long and difficult process.

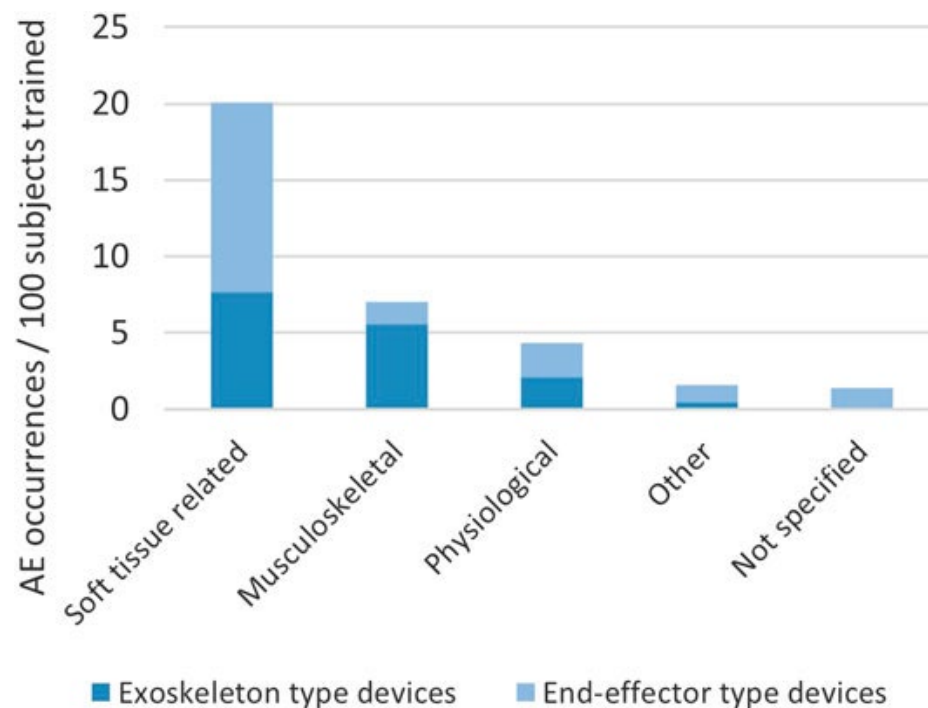
挑战之三 用户安全

Challenges and future trends

● Challenge 3: Ensuring user safety

柱状图：固定式上肢机器人训练中不良事件（AE）发生率/每100受训人。

注意软组织相关事件中末端执行器型比外骨骼型还高——因为末端执行器只抓手，手臂其他部分失控时容易扭伤。



- 软组织相关事件最多（外骨骼型约8次/100人，末端执行器型额外约12次）
- 肌肉骨骼损伤
- 生理事件
- 其他

Types and occurrence of adverse events (AE) in stationary robotic upper limb training.

确保用户安全是无人监护/远程监护下社区或居家康复训练的关键考量。

- Ensuring user safety is a key consideration for unsupervised or remotely supervised community-based or home-based therapy training.

未来趋势1:

临床医生与工程师协作

Challenges and future trends

● Future trends 1: Collaborations between clinicians and exoskeleton designers

协作：通过小样本靶向案例研究理解生物机制，支撑大规模临床试验，驱动功能结局改善，设计个体化助力优化算法。



- **Collaborations:** will enable targeted case studies with a small number of selected participants to inform biological mechanisms that warrant larger clinical trials and drive changes in functional outcomes, to enable the design of optimization algorithms for individualized assistance.

2. 居家康复

Challenges and future trends

● Future trends 2: Home-Based Rehabilitation Solutions

居家：开发可在家中使用的康复机器人是增长方向。需要更便携、更易用的设备，患者可独立使用，增加康复服务可及性，促进持续治疗。

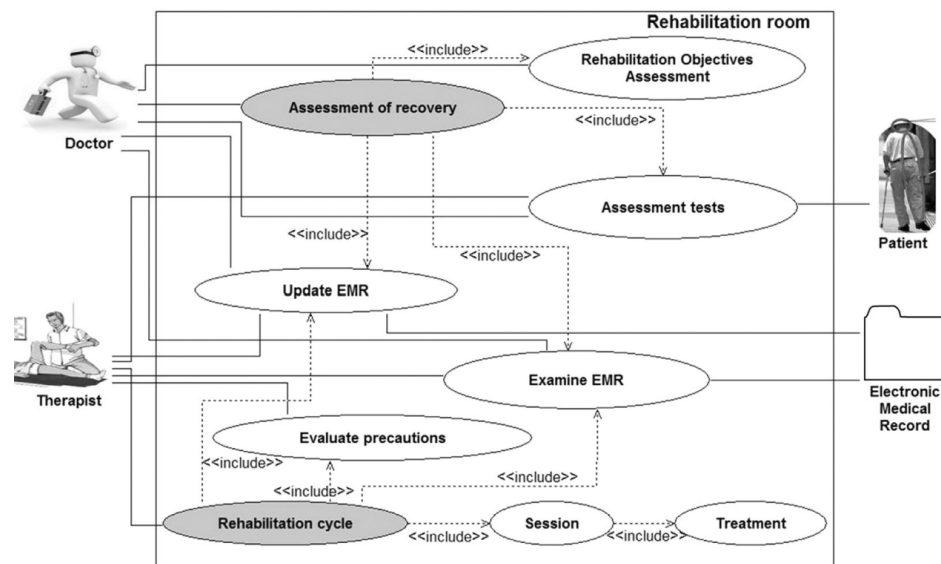


(Home-based upper limb rehabilitation robots)

- **Home-based:** is a growing interest in developing rehabilitation robotics that can be used in a home setting. This could involve more portable and user-friendly devices that patients can use independently, increasing access to rehabilitation services and promoting consistent therapy.

Challenges and future trends

● Future trends 3: Connection between Large Language Models and exoskeletons



[Unified Modeling Language (UML)]

LLM在外骨骼中：AI让大语言模型成为人与可穿戴机器人之间的重要桥梁。把人类指令翻译成机器能懂的语言，把机器反馈翻译成人能懂的语言。这种交互方式大幅提升机器人易用性，让它们更好融入工作和生活。

- **LLM in exoskeletons:** AI makes large language models an important bridge between humans and wearable robotics. They can translate human commands into a language that machines understand and convert machine feedback into a language comprehensible to humans. This interactive approach greatly enhances the usability of robot, allowing them to better integrate into our work and life.